

Analysts' Forecasts: An Empirical Analysis

Bachelor's Thesis

Submitted to

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Degree program: B.Sc. Economics and Business Administration

Submission date: March 25, 2026

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List of Abbreviations

AR(1)	First-Order Autoregressive Process
CF	Cash Flow News
CFO	Chief Financial Officer
CRSP	Center for Research in Security Prices
DR	Discount Rate News
DPS	Dividends Per Share
EPS	Earnings Per Share
ETF	Exchange-Traded Fund
G-H	Graham-Harvey
I/B/E/S	Institutional Brokers' Estimate System
LTG	Long-Term Growth
MLE	Maximum Likelihood Estimation
OLS	Ordinary Least Squares
S&P	Standard & Poor's
SPY	SPDR S&P 500 ETF
WACC	Weighted Average Cost of Capital

1. Introduction

What drives stock prices? In the present value framework, any investor who prices a stock sets its value equal to the expected discounted stream of future dividends. Any price movement therefore must reflect either a change in expected future cash flows or a change in expected returns (De La O & Myers, 2021). The challenge lies in identifying and quantifying the relative importance of these two channels empirically. If prices move because investors revise their cash flow expectations upward, the price-dividend ratio rises alongside expected dividend growth. If instead prices move because investors require lower returns, the same ratio rises without any change in cash flow expectations. The variance decomposition framework applies this logic by asking how much of the observed variation in price ratios can be attributed to movements in cash flow expectations and in return expectations.

Under rational expectations, variance decompositions based on realized historical data find that a high price-dividend ratio today predicts low future returns but not high future dividend growth. This finding implies that most variation in price ratios reflects time-varying expected returns rather than movements in cash flow expectations (Campbell & Shiller, 1988b; Cochrane, 2008, 2011). These results have motivated a large class of asset pricing models that reproduce return predictability through time-varying required returns and assign a minor role to cash flow expectations. Examples include habit formation (Campbell & Cochrane, 1999), long-run risk (Bansal & Yaron, 2004), and rare disaster models (Barro, 2006; Rietz, 1988).

A newer strand of research challenges this consensus by measuring subjective investor expectations directly from survey data instead of inferring them from realized outcomes under the rational expectations assumption. Greenwood and Shleifer (2014) document that return expectations are rising in market upswings and falling in downturns and thus contradict the rational benchmark under which investors should expect low returns when valuations are high. Building on this, De La O and Myers (2021) construct subjective cash flow growth expectations from I/B/E/S (Institutional Brokers' Estimate System) analyst forecasts. I/B/E/S collects earnings and dividend forecasts from professional brokerage and independent analysts, providing a direct measure of market participants' subjective expectations about future cash flows (De La O & Myers, 2021). De La O and Myers (2021) find that cash flow expectations, as opposed to discount rate variation, account for the vast majority of price ratio movements. Their variance decomposition attributes at least 93% of price-dividend ratio fluctuations and at least 63% of price-earnings ratio fluctuations over 2003–2015 to cash flow news, whereas survey-based return expectations contribute little. These findings directly contradict the rational-expectations consensus and establish subjective cash flow expectations as the dominant driver of price ratio fluctuations. Importantly, De La O and Myers (2021) stress that their decomposition holds for any set of subjective expectations and does not require rationality. It remains open, however, whether these expectations exhibit systematic deviations from rationality such as persistent over- or underoptimism.

Subsequent work answers this question: Beliefs are systematically distorted, with the direction of distortion varying between short-term horizons of one to two years and long-term horizons of three to five years. Bouchaud et al. (2019) document that short-term earnings expectations at the firm level are sticky and underreact to news. In contrast, Bordalo et al. (2019, 2024) show that long-term earnings growth expectations, referred to as LTG, exhibit systematic overreaction. Their evidence indicates that high aggregate LTG predicts lower future stock returns, upward revisions are followed by predictable disappointment in realized earnings growth, and the eventual correction of overly optimistic beliefs produces return reversals. More recent work extends this belief-based perspective to the cross-section. Bordalo et al. (2025) show that Expectations-Based Returns constructed from analyst forecast errors and revisions account for a large share of cross-sectional return spreads, including value, investment, size, and momentum. At the same time, Nagel (2024) raises the concern that survey expectations may be re-engineered from current prices and capture time-varying required returns rather than independent beliefs. In this case, LTG may not reliably serve as a proxy for subjective expectations.

Despite rapid progress, several questions remain open. First, the variance decomposition of De La O and Myers (2021) is based on a relatively short sample ending in 2015 that coincides with unusually high earnings volatility (Nagel, 2024). This raises the question of whether the dominance of cash flow expectations persists when the sample is extended beyond 2015. Second, the role of the forecast horizon is not fully understood. Short-term expectations appear to drive price ratios, while only long-term expectations predict returns. This suggests that expectations at different horizons may serve distinct functions in asset pricing (Adam & Nagel, 2022; Bordalo et al., 2024; De La O & Myers, 2021). Third, the interpretation of LTG as a measure of belief distortions rather than a reflection of time-varying required returns remains subject to debate (Nagel, 2024).

This paper addresses these questions using quarterly S&P 500 data constructed from the Center for Research in Security Prices (CRSP), Compustat, I/B/E/S and the Graham-Harvey Global Business Outlook Survey. First, the analysis replicates and extends the variance decompositions of De La O and Myers (2021) through 2023. Cash flow expectations remain an important driver of price ratio variation. However, the contribution of short-term growth expectations declines substantially when the sample is extended beyond 2015. Then, the paper examines return predictability across forecast horizons following Bordalo et al. (2024), focusing on earnings expectations. Long-term earnings growth expectations are significantly associated with lower aggregate returns at three- and five-year horizons, while short-term earnings growth expectations show little predictive power. The return predictability of LTG is robust to the inclusion of price ratio controls, though the strength of this result varies across price ratio specifications and sample periods. Lastly, the analysis applies the Coibion-Gorodnichenko framework (Coibion & Gorodnichenko, 2015) to examine rationality in long-term expectations. The Coibion-Gorodnichenko framework tests whether forecast errors are predictable from past forecast revisions. Predictability would indicate

systematic over- or underreaction in expectations. Across multiple forecast horizons and revision lags, upward LTG revisions tend to be followed by earnings outcomes that fall below the revised forecasts, which is consistent with overreaction.

Overall, the results reveal a horizon-dependent pattern. Short-term expectations explain much of price ratio variation but show little robust predictive power for future returns. Long-term expectations, by contrast, weakly co-move with price ratios yet are associated with lower future returns at longer horizons and exhibit evidence of overreaction. The pattern is in line with the argument that short-term expectations mainly track current earnings dynamics (Adam & Nagel, 2022; Bordalo et al., 2024) and that long-term expectations reflect systematic belief distortions that temporarily inflate prices (Bordalo et al., 2019, 2024). The concern of Nagel (2024), whether LTG ultimately captures subjective beliefs or mechanically reflects time-varying required returns, cannot be resolved with aggregate data alone.

The paper is organized as follows. Section 2 develops the methodological framework. Section 3 describes the data and construction of variables. Section 4 presents the variance decomposition results. Section 5 examines return predictability across forecast horizons. Section 6 tests forecast rationality via the Coibion-Gorodnichenko regression and Section 7 concludes.

2. Methodology

This section develops the analytical frameworks employed in the empirical analysis: the variance decomposition of De La O and Myers (2021) and the Coibion-Gorodnichenko regression (Coibion & Gorodnichenko, 2012, 2015).

2.1 Variance Decomposition

By definition, a stock price equals the discounted value of future dividends. When looking at an index, like the S&P 500, its value is the discounted value of future dividends paid by all firms listed in the index. If the level of the index moves, then it must reflect changes in expected future cash flows or changes in discount rates (De La O & Myers, 2021). The variance decomposition formalizes this identity and allows to quantify empirically which channel dominates. Following De La O and Myers (2021), this paper begins by deriving the decomposition for the price-dividend ratio, then extends it to the price-earnings ratio by applying the payout ratio identity, which links dividends and earnings through the dividend-to-earnings ratio. The paper also adopts the notational convention of marking operators with an asterisk when computed under the subjective probability distribution, and leaving them unmarked when computed under the objective distribution. Accordingly, $E^*[\cdot]$ denotes a subjective expectation, whereas $\text{cov}(\cdot, \cdot)$ denotes the objective covariance.

The analysis is based on the identity that stock prices must equal the present value of expected future cash flows. For the aggregate S&P 500 index, this relationship can be expressed through

the one-year return identity:

$$R_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t} = \frac{\left(\frac{P_{t+1}}{D_{t+1}} + 1\right) \frac{D_{t+1}}{D_t}}{\frac{P_t}{D_t}}, \quad (1)$$

where R_{t+1} is the gross return, P_t refers to the price level, and D_t to the dividends of the index at time t (De La O & Myers, 2021).

Following Campbell and Shiller (1988a), log-linearizing this identity around the long-run average price-dividend ratio yields the relationship of the price-dividend ratio pd_t determined by future dividend growth, Δd_{t+1} , future returns, r_{t+1} , and the subsequent price-dividend ratio pd_{t+1} :

$$pd_t = \kappa + \Delta d_{t+1} - r_{t+1} + \rho pd_{t+1}, \quad (2)$$

where lowercase variables denote natural logarithms, κ is a linearization constant, and $\rho = \frac{e^{\overline{pd}}}{1+e^{\overline{pd}}} < 1$ is the discount factor determined by the mean log price-dividend ratio $\overline{pd}$, which is estimated at $\rho = 0.977$ from the sample mean of the log price-dividend ratio. By iterating the log-linear equation forward and imposing the no-bubble condition $\lim_{T \rightarrow \infty} \rho^T E_t^*[pd_{t+T}] = 0$, it follows that

$$pd_t = \frac{\kappa}{1-\rho} + \sum_{j=1}^{\infty} \rho^{j-1} E_t^*[\Delta d_{t+j}] - \sum_{j=1}^{\infty} \rho^{j-1} E_t^*[r_{t+j}], \quad (3)$$

where $E_t^*[\cdot]$ denotes subjective expectations formed by market participants at time t . This equation shows that the current price-dividend ratio equals a constant plus two components: (i) cash flow expectations through the present value of expected dividend growth, and (ii) discount rate expectations through the present value of expected returns. This decomposition holds under any subjective probability distribution, regardless of whether expectations are rational (De La O & Myers, 2021).

Equation (3) shows that any movement in the price-dividend ratio must originate from one of two sources: a shift in subjective dividend growth expectations or a change in subjective expected returns. While this identity holds by construction, it does not indicate which channel dominates empirically. Following the variance decomposition of Campbell and Shiller (1988a), the relative contribution of each channel is quantified by taking the covariance of equation (3) with pd_t and dividing by $\text{var}(pd_t)$:

$$1 = \underbrace{\frac{\text{cov}\left(\sum_{j=1}^{\infty} \rho^{j-1} E_t^*[\Delta d_{t+j}], p d_t\right)}{\text{var}(p d_t)}}_{\text{Cash Flow News (CF)}} + \underbrace{\frac{\text{cov}\left(-\sum_{j=1}^{\infty} \rho^{j-1} E_t^*[r_{t+j}], p d_t\right)}{\text{var}(p d_t)}}_{\text{Discount Rate News (DR)}}. \quad (4)$$

The cash flow news (CF) measures the fraction of price-dividend ratio variance explained by changes in subjective dividend growth expectations. The discount rate news (DR) measures the fraction explained by changes in subjective return expectations. By construction, $\text{CF} + \text{DR} = 1$. Furthermore, based on De La O and Myers (2021), the same logic extends to the price-earnings ratio using the log payout ratio $de_t = \log(D_t/E_t)$. Substituting the identity $pe_t = p d_t + de_t$ into equation (2) and rearranging yields:

$$pe_t = \kappa + \Delta e_{t+1} - r_{t+1} + (1 - \rho) de_{t+1} + \rho pe_{t+1}. \quad (5)$$

Since $(1 - \rho) = 0.023$, movements in the expected future payout ratio have minimal impact on the price-earnings ratio. This term is therefore treated as negligible, giving:

$$pe_t \approx \tilde{\kappa} + \Delta e_{t+1} - r_{t+1} + \rho pe_{t+1} \quad (6a)$$

$$\approx \frac{\tilde{\kappa}}{1 - \rho} + \sum_{j=1}^{\infty} \rho^{j-1} E_t^*[\Delta e_{t+j}] - \sum_{j=1}^{\infty} \rho^{j-1} E_t^*[r_{t+j}], \quad (6b)$$

where $\tilde{\kappa} = \kappa + (1 - \rho)\overline{de}$ and $\overline{de}$ denotes the mean log payout ratio. This results in a variance decomposition for the price-earnings ratio analogous to equation (4), where cash flow news now reflects earnings growth expectations instead of dividend growth expectations:

$$1 \approx \frac{\text{cov}\left(\sum_{j=1}^{\infty} \rho^{j-1} E_t^*[\Delta e_{t+j}], pe_t\right)}{\text{var}(pe_t)} + \frac{\text{cov}\left(-\sum_{j=1}^{\infty} \rho^{j-1} E_t^*[r_{t+j}], pe_t\right)}{\text{var}(pe_t)}. \quad (7)$$

Analogously, fluctuations in the price-earnings ratio must reflect either shifts in subjective earnings growth expectations or in subjective expected returns (De La O & Myers, 2021).

2.2 Coibion-Gorodnichenko Regression Analysis

Drawing on the framework of Coibion and Gorodnichenko (2012, 2015), the paper tests forecast rationality by examining whether forecast errors are predictable from forecast revisions. Under rationality, the forecast revision should be uncorrelated with future forecast errors, since all available information is already incorporated into the forecast. A positive correlation points to underreaction, where forecasters update too conservatively; a negative correlation points to

overreaction, where forecasters overshoot in response to news.

Based on Bouchaud et al. (2019) and Coibion and Gorodnichenko (2015), potential deviations from the rational benchmark are interpreted through the sticky-information model of Mankiw and Reis (2002). This model assumes that forecasters update their information sets with probability $1 - \lambda$ in each period, where λ captures the degree of information rigidity. Agents who update their information sets are assumed to acquire full information and form rational expectations (Coibion & Gorodnichenko, 2015).

Under sticky information, the consensus forecast at time t can be expressed as:

$$E_t^*[x_{t+h}] = (1 - \lambda) \sum_{j=0}^{\infty} \lambda^j E_{t-j}[x_{t+h}], \quad (8)$$

where $E_t^*[x_{t+h}]$ is the average forecast at time t for variable x at time $t + h$, and $E_{t-j}[x_{t+h}]$ represents the full-information rational expectation formed at time $t - j$. This structure indicates that the current consensus forecast is a weighted average of current and past full-information rational expectations, with exponentially declining weights on older information sets.

The key insight of Coibion and Gorodnichenko (2015) is that sticky information generates a specific predictable relationship between forecast errors and forecast revisions. By manipulating the sticky-information equation, they derive:

$$x_{t+h} - E_t^*[x_{t+h}] = \frac{\lambda}{1 - \lambda} \left(E_t^*[x_{t+h}] - E_{t-1}^*[x_{t+h}] \right) + \varepsilon_{t+h,t}, \quad (9)$$

where $\varepsilon_{t+h,t}$ is the rational expectations forecast error, which is orthogonal to all information available at time t or earlier.

This equation reveals that the coefficient on the forecast revision, $\beta = \frac{\lambda}{1 - \lambda}$, provides a direct measure of information rigidity. Under full-information rational expectations, forecast errors are unpredictable from forecast revisions, yielding $\beta = 0$. Under information rigidities, forecasters underreact to new information, so positive forecast revisions predict positive forecast errors and $\beta > 0$. Under overreaction, forecasters overshoot, so positive forecast revisions predict negative forecast errors and $\beta < 0$ (Bouchaud et al., 2019; Coibion & Gorodnichenko, 2015).

3. Data and Construction of Variables

This section describes the data used in the empirical analysis and explains how the main variables are constructed. Following De La O and Myers (2021), the paper builds realized S&P 500 measures for dividends, earnings, and returns, as well as the corresponding forecast indices.

3.1 S&P 500 Realized Cash Flow Index

The analysis begins by constructing a quarterly panel of S&P 500 constituents using CRSP. For each quarter, the set of firms belonging to the index at that point in time is identified, including firms that enter or leave the index. Quarterly firm-level fundamentals are gathered from Compustat for all firms in the S&P 500, including prices, shares outstanding, and earnings per share (EPS).¹ In addition, dividends per share (DPS) are obtained from CRSP, which allows the analysis to focus on ordinary cash dividends excluding special items.

Aggregate index-level dividends and earnings are constructed by first computing total dividends and total earnings at the firm level as trailing four-quarter sums. These are then summed across all S&P 500 constituents within each quarter and divided by an index divisor, constructed at quarterly frequency. One-year realized growth rates are computed as the log difference between the aggregate index four quarters ahead and its current level. Since the analysis uses data from S&P 500 firms, aggregate dividends and earnings are positive in each quarter, which ensures that log differences are well-defined. Details on the construction of the indices are provided in the Appendix.

3.2 S&P 500 Subjective Cash Flow Expectations Index

In the following, the paper constructs three expectation series from I/B/E/S analyst forecasts: short-term one- and two-year ahead expectations, and a long-term growth index, each aggregated to the index level.

3.2.1 One- and Two-Year Expectations

Forecast data are drawn from the I/B/E/S Unadjusted US Summary Statistics file. I/B/E/S provides consensus forecasts at multiple horizons for each firm listed in the United States. As noted by De La O and Myers (2021), I/B/E/S collects non-anonymous forecasts from brokerage and independent analysts; because estimates are attributed to individual analysts or firms, reputational and career incentives encourage accurate reporting. Aggregate expectations at the index level are obtained by combining firm-level forecasts for the S&P 500 constituents in each period, following the same procedure used to construct the realized index. Implementation details are provided in the Appendix.

From I/B/E/S, EPS and DPS median forecasts are downloaded for up to five annual fiscal periods ahead. EPS one-year expectations are available from 1976Q1, two-year expectations

¹De La O and Myers (2021) do not explicitly specify which EPS definition they use. It is assumed that they use earnings per share excluding extraordinary items (epspxq in Compustat). This assumption is supported by the replication results: Using earnings including extraordinary items reduces the correlation between the replicated and the original one-year earnings growth expectation series of De La O and Myers (2021) in Table 2 from 0.946 to 0.828, whereas the excluding-extraordinary-items definition yields the closer match.

from 1985Q1, and DPS one- and two-year expectations from 2003Q1 (De La O & Myers, 2021). For each series, two samples are constructed: a replication subsample running from the earliest available date through 2015Q3, enabling direct comparison with De La O and Myers (2021), and an extended sample continuing through 2023Q3 to capture more recent developments in investor expectations.

Before constructing aggregate cash flow expectations, two aspects of the I/B/E/S data must be addressed. First, forecasts are issued for fiscal periods anchored to each firm's fiscal year-end and not to a fixed calendar horizon. Adjacent fiscal-year forecasts are therefore interpolated to obtain exact one- and two-year ahead estimates for each firm, following the procedure of De La O and Myers (2021). Second, I/B/E/S does not report forecasts for every S&P 500 constituent because some firms lack analyst coverage. To address this coverage gap, aggregate dividend and earnings expectations constructed from covered firms are scaled by the ratio of total S&P 500 market capitalization to the market capitalization of the forecast-covered subset. This scaling relies on the assumption that covered firms form a representative sample of the index.

A robustness check assesses the plausibility of this assumption using realized cash flows. Realized S&P 500-level dividend and earnings series are constructed from all index constituents and separately from only the forecast-covered firms, rescaled by the market-capitalization ratio. If covered firms are representative, the rescaled realized series should closely track the full-constituent series. The aggregate series are additionally benchmarked against Robert Shiller's S&P 500 dividend and earnings data (Shiller, 2015) and the dividend series implied by SPY (an S&P 500 tracking ETF from CRSP).

Table 1 presents these correlations for dividend and earnings levels and growth rates across both panels. The dividend and earnings series constructed from all S&P 500 firms (*All Companies*) are highly correlated with the external benchmarks (*Shiller*, *SPY*), supporting the accuracy of the aggregation method. Moreover, the *I/B/E/S* series remain highly correlated with the *All Companies* series and the external benchmarks, indicating that firms with analyst coverage are representative of the broader index. Moving from levels to growth rates yields the same conclusion. Together, the evidence supports the validity of the scaling assumption. The correlations closely match those reported by De La O and Myers (2021); in the Appendix, the validation is repeated for their original subsample ending in 2015Q3 and the correlations are compared directly to theirs.

As a final step, quarterly time series of aggregate expected dividend and earnings growth are constructed by taking log differences of the corresponding level forecasts. To validate the implementation, the one-year-ahead dividend and earnings growth series are compared against the original data provided by De La O and Myers (2021), constructed following their methodology and made publicly available online.² Since both series are based on the same construction procedure,

²The replication data are available at <https://www.ricardodelao.com/data>.

any discrepancies reflect differences in data sources or retrieval rather than methodological choices, and close agreement between the two series confirms the accuracy of the replication.

Table 1: Correlations of S&P 500 Cash Flow Measures

	Levels			Growth		
	I/B/E/S (1)	Shiller (2)	SPY (3)	I/B/E/S (4)	Shiller (5)	SPY (6)
<i>Panel A: Dividends — 2004Q1–2023Q3 (N = 79 levels, N = 75 growth)</i>						
All Companies	0.999	0.997	0.997	0.969	0.965	0.930
I/B/E/S		0.998	0.998		0.949	0.905
Shiller			0.999			0.966
<i>Panel B: Earnings — 1977Q1–2023Q3 (N = 187 levels, N = 183 growth)</i>						
All Companies	0.997	0.997		0.993	0.926	
I/B/E/S		0.998			0.919	
Shiller						

Note: The table reports pairwise correlations for quarterly S&P 500 cash flow series. *All Companies* aggregates realized cash flows across all index constituents; *I/B/E/S* aggregates realized cash flows only for constituents with analyst coverage, rescaled by the market-capitalization ratio; *Shiller* refers to the quarterly S&P 500 series from Shiller (2015). Panel A additionally benchmarks against SPY (SPDR S&P 500 ETF, CRSP). Panel B uses three series only (no SPY earnings equivalent). The growth columns report correlations of log annual changes.

Table 2 reports that the correlation between the replicated and the original series is 0.878 for expected dividend growth and 0.946 for expected earnings growth. The comparison is conducted over the period for which the original data are available, ending in 2022Q1. Overall, these results confirm that the implementation closely replicates the original methodology of De La O and Myers (2021). Remaining differences likely reflect minor variations in data sources or retrieval.

Table 2: Replication Validation: Correlation with De La O & Myers (2021) Data

Series	Correlation	Sample window	N
Dividend growth	0.878	2003Q1–2022Q1	77
Earnings growth	0.946	1976Q1–2022Q1	185

Note: This table compares the replicated one-year-ahead expected cash flow growth series against the original series provided by De La O and Myers (2021), available at <https://www.ricardodelao.com/data>. The growth rate is computed as the log annual change of the corresponding level forecast. The sample window is limited to the period covered by the original data. Correlations are computed over the overlapping sample.

Two-year dividend and earnings level and growth forecasts are constructed following the identical procedure as their one-year counterparts, including interpolation between annual forecast horizons and market-capitalization-weighted scaling.

3.2.2 Long-Term Growth Forecasts

Following Bordalo et al. (2024), this paper also builds long-term growth (LTG) forecast indices for dividends and earnings. LTG forecasts are downloaded from the I/B/E/S US Unadjusted

Summary Statistics table, available from December 1981 onward, using the consensus (median) forecast at the firm level. I/B/E/S describes LTG as the expected average annual growth over three to five years, spanning a firm’s next full business cycle (Bordalo et al., 2019). These firm-level LTG forecasts are then aggregated to the index level using value weights. This yields a quarterly time series of long-term growth expectations for the S&P 500. Implementation details are provided in the Appendix. While EPS LTG coverage is stable throughout the sample, I/B/E/S discontinued systematic collection of DPS LTG forecasts after mid-2015, with fewer than five S&P 500 constituents reporting thereafter. The DPS LTG series is therefore restricted to 2003Q1–2015Q3. Figure A2 in the Appendix documents this.

3.2.3 Characteristics of Analyst Forecast Series

Table 3 shows annualized standard deviations, correlations, and co-movement statistics for all expectation series with the price-dividend and price-earnings ratios. Two sample windows are reported: the full available sample for each series and the replication subsample of De La O and Myers (2021) ending in 2015Q3.

Panels A and B show that short-term subjective cash flow growth expectations are time-varying. One-year earnings growth has a standard deviation of 22.2% over the full available 1976Q1–2023Q3 sample, rising to 25.1% when the sample is restricted to start in 2003Q1, and further to 31.1% in the replication subsample of 2003Q1–2015Q3. The progressive increase reflects the exceptional earnings volatility of the 2003–2015 period, which is diluted once post-2015 observations are included. The co-movement with the price-earnings ratio follows the same pattern: The correlation rises from 0.705 in the full available sample to 0.806 and 0.921 in the two shorter windows, and the co-movement coefficient increases correspondingly from 0.378 to 0.761 and 0.935. In the replication subsample, one-year earnings growth expectations thus account for nearly all price-earnings ratio variation – a finding Section 4 revisits in the variance decomposition.

Dividend growth expectations are smoother, with standard deviations of 6.3% and 6.0% for the one- and two-year horizons over the full available 2003Q1–2023Q3 sample. Their co-movement with the price-dividend ratio displays the same sample-dependence: The correlation for one-year dividend growth rises from 0.498 in the full available sample to 0.813 in the replication subsample, with the co-movement coefficient increasing from 0.196 to 0.394. Two-year dividend growth exhibits a comparable pattern, with co-movement rising from 0.161 to 0.334. The results of earnings and dividend growth expectations indicate that a share of price ratio movements is associated with short-term cash flow growth expectations, though the strength of this link is more pronounced in the replication subsample and weakens considerably once the post-2015 data are included.

LTG expectations are the smoothest series overall, with standard deviations of just 2.1% for

earnings LTG and 3.7% for dividend LTG in their respective full available samples. Their co-movement with price ratios is close to zero: 1.7% for earnings LTG and 1.0% for dividend LTG. The same qualitative pattern holds in the replication subsample. These results suggest that price ratios are primarily driven by short-run growth expectations, while LTG is largely orthogonal to contemporaneous price ratio movements.

Table 3: Co-Movement of Subjective Cash Flow Expectations and Price Ratios

Series	Horizon	Sample	Std. Dev.	Corr.	$\frac{\text{cov}(\cdot, pd_t)}{\text{var}(pd_t)}$	$\frac{\text{cov}(\cdot, pe_t)}{\text{var}(pe_t)}$
			(1)	(2)	(3)	(4)
<i>Panel A: Dividend Growth Expectations</i>						
I/B/E/S 1yr	1	2003Q1–2023Q3	6.3%	0.498	0.196 (0.079)	
		2003Q1–2015Q3	7.5%	0.813	0.394 (0.054)	
I/B/E/S 2yr	2	2003Q1–2023Q3	6.0%	0.429	0.161 (0.070)	
		2003Q1–2015Q3	7.0%	0.742	0.334 (0.064)	
I/B/E/S LTG	LTG	2003Q1–2015Q3	3.7%	0.042	0.010 (0.049)	
<i>Panel B: Earnings Growth Expectations</i>						
I/B/E/S 1yr	1	1976Q1–2023Q3	22.2%	0.705		0.378 (0.082)
		1976Q1–2015Q3	24.0%	0.740		0.409 (0.087)
		<i>2003Q1–2023Q3</i>	<i>25.1%</i>	<i>0.806</i>		<i>0.761 (0.198)</i>
		<i>2003Q1–2015Q3</i>	<i>31.1%</i>	<i>0.921</i>		<i>0.935 (0.123)</i>
I/B/E/S 2yr	2	1985Q1–2023Q3	12.2%	0.663		0.256 (0.070)
		1985Q1–2015Q3	13.4%	0.699		0.272 (0.073)
I/B/E/S LTG	LTG	1981Q4–2023Q3	2.1%	0.282		0.017 (0.010)
		1981Q4–2015Q3	1.6%	0.363		0.015 (0.012)
<i>Panel C: Return Expectations (G-H CFO Survey)</i>						
G-H	1	2003Q1–2023Q3	1.4%	0.469	0.039 (0.012)	0.004 (0.007)
		2003Q1–2015Q3	1.4%	0.641	0.059 (0.017)	0.004 (0.010)
G-H	10	2003Q1–2023Q3	0.7%	0.358	0.015 (0.010)	0.007 (0.003)
		2003Q1–2015Q3	0.7%	0.353	0.017 (0.015)	0.010 (0.005)

Note: This table presents annualized standard deviations, Pearson correlations, and co-movement statistics for subjective cash flow and return expectation series with the S&P 500 price-dividend ratio (pd_t) and price-earnings ratio (pe_t). Panels A and B report I/B/E/S-based dividend and earnings growth expectations at one-year, two-year, and long-term (LTG) horizons. The two-year horizon series represents the annualized cumulative growth expectation over two years, computed as $\frac{1}{2}(\log E_t^*[CF_{t+2}] - \log CF_t)$, based on De La O and Myers (2021). Panel C reports one- and ten-year return expectations from the Graham-Harvey CFO survey (G-H). Column (1) reports the annualized standard deviation of each series. Column (2) reports the Pearson correlation between each series and the relevant price ratio: $\text{Corr}(\cdot, pd_t)$ for Panel A and Panel C, and $\text{Corr}(\cdot, pe_t)$ for Panel B. Columns (3) and (4) report the covariance of each series with the respective price ratio, normalized by the ratio's variance. Small-sample adjusted Newey-West standard errors are reported in parentheses. For the I/B/E/S one-year earnings series in Panel B, two additional rows (in italics) report the same statistics restricted to start in 2003Q1, matching the variance decomposition sample window used in Section 4.

Figure 1 illustrates the same pattern visually: One- and two-year forecasts swing sharply over the business cycle and LTG evolves more gradually. This suggests that LTG captures lower-frequency movements in aggregate investor optimism rather than transitory earnings dynamics. Bordalo

et al. (2024) document the same pattern and highlight that it is precisely the slow-moving nature of LTG that gives it predictive power for multi-year stock returns, a mechanism Section 5 revisits.

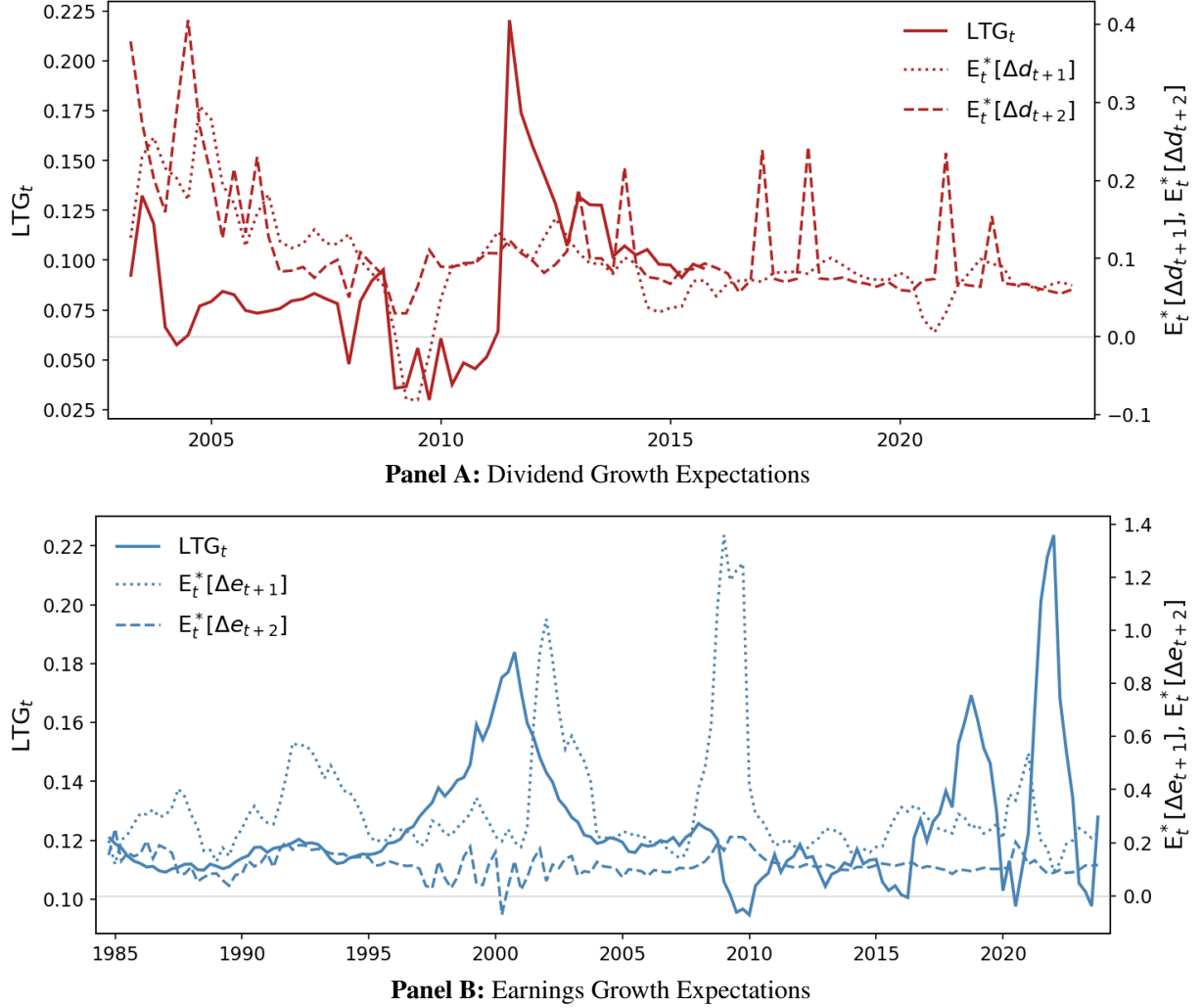


Figure 1: S&P 500 Cash Flow Growth Expectations: LTG, One-Year Ahead, and Two-Year Ahead

Note: This figure plots quarterly time series of aggregate S&P 500 cash flow growth expectations constructed from I/B/E/S data, expressed as decimals. The solid line shows the long-term growth (LTG) forecast index (left axis). The dotted and dashed lines show the one-year-ahead and two-year-ahead expected growth rates (right axis). Panel A shows one- and two-year dividend growth expectations from 2003Q1 to 2023Q3 and dividend LTG is available only through 2015Q3. Panel B covers earnings growth expectations from 1985Q1 to 2023Q3.

The co-movement coefficients and standard deviations reported in Table 3 replicate those of De La O and Myers (2021) for the overlapping 2003Q1–2015Q3 subsample. Minor differences are likely attributable to small discrepancies in the underlying expectation series, as evidenced by the replication correlations of 0.946 for earnings growth and 0.878 for dividend growth reported in Table 2.

3.3 S&P 500 Realized Returns

The paper constructs aggregate S&P 500 realized returns from monthly index returns obtained from CRSP. After converting to log returns, these are aggregated to a quarterly frequency to

match the frequency of the I/B/E/S forecast data. Forward-looking realized returns over one, three, and five years are constructed: The one-year return is the cumulative log return over the subsequent four quarters, while the three- and five-year returns are discounted cumulative returns of the form $\sum_{j=1}^h \rho^{j-1} r_{t+j}$, where each r_{t+j} denotes the annual log return in year j ahead and ρ is the log-linearization discount factor defined in Section 2.

3.4 S&P 500 Subjective Return Expectations

Based on De La O and Myers (2021), the paper measures subjective return expectations using the Graham-Harvey CFO Survey, conducted by John Graham and Campbell Harvey of Duke University. The survey collects expected S&P 500 returns over the next 12 months and over the next 10 years from 200 to 500 CFOs of U.S. corporations (De La O & Myers, 2021). The data are publicly available online³ and cover the period from 2001Q4 onward at a quarterly frequency. Specifically, the analysis uses the mean expected return for the pre-2020 period and the median expected return from 2020 onwards, reflecting a change in the variable definitions across the two data vintages available on the survey website. As documented in Panel C of Table 3, all measures of subjective return expectations exhibit low standard deviations relative to cash flow growth expectations and display small co-movement with the price-dividend and price-earnings ratio. This pattern is also present in the subsample ranging until 2015Q3 and aligns with the findings of De La O and Myers (2021). These results suggest that subjective return expectations account for only a limited share of variation in price ratios.

After constructing all series, including subjective cash flow expectations, return expectations, and realized measures for the S&P 500, the next section applies the variance decomposition of De La O and Myers (2021) to quantify the relative importance of cash flow and return expectations in driving movements in the price-dividend and price-earnings ratios.

4. Subjective Variance Decomposition of the S&P 500 Price Ratios

This section implements the variance decomposition of De La O and Myers (2021) to estimate the share of price ratio fluctuations attributable to cash flow news and discount rate news. Two sample periods are considered: a replication subsample from 2003Q1 to 2015Q3, enabling direct comparison with De La O and Myers (2021), and an extended sample from 2003Q1 to 2023Q3 that incorporates the post-2015 period. The analysis proceeds in three steps: A finite-horizon decomposition using one-year expectations is first estimated to assess the relative importance of short-run cash flow and return expectations. Then, the decomposition is extended to include two-year cash flow growth expectations and ten-year return expectations. Lastly, all available horizons are combined to evaluate the joint contribution of subjective cash flow expectations relative to subjective return expectations.

³<https://www.cfosurvey.org>

4.1 One-Year Horizon Decomposition

The analysis begins with the short-term horizon, in line with De La O and Myers (2021). Taking expectations of equation (2) implies that variation in the price-dividend ratio can arise from changes in expected one-year dividend growth, expected one-year returns, or changes in the expected future price-dividend ratio. Formally:

$$1 = \underbrace{\frac{\text{cov}(E_t^*[\Delta d_{t+1}], pd_t)}{\text{var}(pd_t)}}_{CF_1} + \underbrace{\frac{\text{cov}(-E_t^*[r_{t+1}], pd_t)}{\text{var}(pd_t)}}_{DR_1} + \rho \underbrace{\frac{\text{cov}(E_t^*[pd_{t+1}], pd_t)}{\text{var}(pd_t)}}_{LT}, \quad (10)$$

where CF_1 and DR_1 measure the fraction of price-dividend ratio variance explained by one-year subjective dividend growth and return expectations, respectively. LT absorbs all expectational components beyond the one-year horizon. One-year subjective dividend growth and return expectations are directly observable from I/B/E/S and the Graham-Harvey survey, whereas the one-year subjective price-dividend ratio expectation $E_t^*[pd_{t+1}]$ is not directly measured but can be recovered by taking expectations of equation (2) and solving for the expected future price-dividend ratio:

$$E_t^*[pd_{t+1}] = \frac{pd_t - \kappa - E_t^*[\Delta d_{t+1}] + E_t^*[r_{t+1}]}{\rho}. \quad (11)$$

The variance decomposition for the price-earnings ratio is derived analogously from equation (6a):

$$1 \approx \underbrace{\frac{\text{cov}(E_t^*[\Delta e_{t+1}], pe_t)}{\text{var}(pe_t)}}_{CF_1} + \underbrace{\frac{\text{cov}(-E_t^*[r_{t+1}], pe_t)}{\text{var}(pe_t)}}_{DR_1} + \rho \underbrace{\frac{\text{cov}(E_t^*[pe_{t+1}], pe_t)}{\text{var}(pe_t)}}_{LT}. \quad (12)$$

De La O and Myers (2021) add that a property of this decomposition is that one-year cash flow news of dividends and earnings, as well as one-year discount rate news, are estimated independently. As a result, return expectations do not influence the cash flow component, and cash flow expectations do not influence the discount rate component. This remains true even if the investors surveyed about cash flow growth and those surveyed about returns hold different beliefs. In that case, LT captures the remaining variation in the price ratio that is not explained by movements in the first group's cash flow growth expectations or the second group's return expectations.

Table 4 reports the one-year variance decomposition results for both price ratios. For the price-dividend ratio, CF_1 equals 0.196 in the extended sample, implying that one-year subjective dividend growth expectations explain 19.6% of price-dividend ratio variation. The discount rate channel is close to zero at -0.037 . The majority of the variation is absorbed by the long-term residual $LT = 0.841$, indicating that most price-dividend ratio fluctuations are driven by

expectations at horizons beyond one year.

Table 4: One-Year Horizon Variance Decomposition of S&P 500 Price Ratios

	Sample (1)	CF_1 (2)	DR_1 (3)	LT (4)
Price-dividend ratio	2003Q1 to 2023Q3	0.196** (0.081)	−0.037*** (0.012)	0.841*** (0.078)
	2003Q1 to 2015Q3	0.394*** (0.054)	−0.056*** (0.016)	0.662*** (0.054)
Price-earnings ratio	2003Q1 to 2023Q3	0.775*** (0.197)	−0.004 (0.007)	0.228 (0.201)
	2003Q1 to 2015Q3	0.935*** (0.123)	−0.004 (0.009)	0.069 (0.132)

Note: Each row decomposes price-ratio variance into a one-year cash flow component (CF_1), a one-year discount rate component (DR_1), and a residual long-horizon term (LT). The three components are estimated by separately regressing the one-year subjective cash flow growth expectation, the one-year subjective return expectation, and ρ times the implied one-year-ahead price ratio on the current log price ratio. By construction, $CF_1 + DR_1 + LT = 1$, so that LT absorbs all expectational variation beyond the one-year horizon. Newey-West standard errors with small-sample correction are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Restricting to the replication subsample raises CF_1 to 0.394 while DR_1 remains close to zero at −0.056. Correspondingly, LT declines to 0.662. These figures closely replicate the benchmark values of De La O and Myers (2021), who report $CF_1 = 0.390$, $DR_1 = -0.049$, and $LT = 0.659$. The higher CF_1 in the replication subsample reflects the tighter link between dividend growth beliefs and market conditions during this period: Investors anticipated stronger dividend growth in market upturns and weaker growth in downturns, causing dividend growth expectations to co-move tightly with the price-dividend ratio (De La O & Myers, 2021). The decline in LT from 0.841 to 0.662 confirms that a comparatively larger fraction of price-dividend ratio movements was tied to near-term beliefs in the replication subsample. The lower CF_1 in the extended sample therefore reflects a weaker link between one-year dividend growth expectations and the price-dividend ratio after 2015. Since dividend growth volatility remained broadly stable (7.5% in the replication subsample vs. 6.3% in the extended sample, see Table 3), this decline is driven by the weaker correlation between dividend growth expectations and price-dividend ratio movements in the post-2015 period (0.813 vs. 0.498, see Table 3).

For the price-earnings ratio, one-year earnings growth expectations account for 77.5% of price-earnings ratio variation in the extended sample, with $DR_1 = -0.004$ and $LT = 0.228$. Narrowing to the replication subsample pushes CF_1 to 0.935 and shrinks LT to 0.069, closely matching the reported values of De La O and Myers (2021) of $CF_1 = 0.937$ and $LT = 0.064$. The near-zero LT implies that almost all price-earnings ratio variation in the replication subsample was linked to investors' short-term earnings beliefs. De La O and Myers (2021) argue that this high co-movement is driven by the exceptional volatility of earnings expectations in that period.

The lower CF_1 of 0.775 in the extended sample reflects a contraction in earnings growth volatility and a weakening of its correlation with the price-earnings ratio after 2015. As documented in the italicized rows of Table 3, the standard deviation of one-year earnings growth expectations falls from 31.1% to 25.1% and the correlation drops from 0.921 to 0.806 across the two windows. The standard deviation of the log price-earnings ratio also contracts, partially offsetting the decline in CF_1 .⁴ Together, the net effect is a decline in CF_1 from 0.935 to 0.775, because the reduction in earnings growth volatility and the weakening of its correlation with the price-earnings ratio outweigh the offsetting effect of the lower price-earnings ratio variance after 2015.

Across both price ratios and both sample windows, DR_1 is close to zero and negative. This finding is consistent with the results from Table 3 that subjective return expectations vary less over time than cash flow beliefs and that most of their variation is orthogonal to price ratios. One-year return expectations from the Graham-Harvey survey thus appear to contribute little to explaining price ratio movements. Instead, price ratio fluctuations are more strongly associated with measured cash flow expectations.

4.2 Extended Decomposition

The analysis is extended to longer horizons by additionally including two-year cash flow expectations and ten-year return expectations, as in De La O and Myers (2021). This allows the paper to assess how much of the long-term residual LT , which by construction captures all expectational components beyond the one-year horizon, can be attributed to short-term expectations just beyond the first year. The extended variance decomposition is computed by regressing the discounted two-year cash flow expectation $\sum_{j=1}^2 \rho^{j-1} E_t^*[\Delta d_{t+j}]$ and the discounted ten-year return expectation $\sum_{j=1}^{10} \rho^{j-1} E_t^*[r_{t+j}]$ on the current log price-dividend ratio pd_t . The resulting slope coefficients define CF_2 and DR_{10} :

$$CF_2 = \frac{\text{cov}\left(\sum_{j=1}^2 \rho^{j-1} E_t^*[\Delta d_{t+j}], pd_t\right)}{\text{var}(pd_t)}, \quad (13)$$

$$DR_{10} = \frac{\text{cov}\left(-\sum_{j=1}^{10} \rho^{j-1} E_t^*[r_{t+j}], pd_t\right)}{\text{var}(pd_t)}. \quad (14)$$

Unlike the one-year decomposition, $CF_2 + DR_{10}$ need not equal one because expectations beyond year two (cash flows) and beyond year ten (discount rates) are not directly measured. Analogously for the price-earnings ratio, the discounted two-year earnings growth expectation

⁴The standard deviation of the log price-earnings ratio is 30.6% in the replication subsample and 26.6% in the extended sample.

$\sum_{j=1}^2 \rho^{j-1} E_t^*[\Delta e_{t+j}]$ and the discounted ten-year return expectation $\sum_{j=1}^{10} \rho^{j-1} E_t^*[r_{t+j}]$ are regressed on the current log price-earnings ratio pe_t , and define the variance decomposition analogously.

Table 5: Extended Horizon Variance Decomposition of S&P 500 Price Ratios

	Sample (1)	CF_2 (2)	DR_{10} (3)
Price-dividend ratio	2003Q1 to 2023Q3	0.316** (0.141)	-0.128 (0.086)
	2003Q1 to 2015Q3	0.661*** (0.126)	-0.147 (0.126)
Price-earnings ratio	2003Q1 to 2023Q3	0.841*** (0.220)	-0.061** (0.028)
	2003Q1 to 2015Q3	1.012*** (0.141)	-0.080* (0.042)

Note: This table extends the one-year decomposition by incorporating two-year cumulative cash flow expectations and ten-year cumulative return expectations, allowing a larger share of price ratio variation to be attributed to directly observed subjective beliefs. CF_2 and DR_{10} are estimated by regressing the discounted two-year cumulative cash flow expectation and the discounted ten-year cumulative return expectation on the current log price ratio, respectively. Specifically, for the price-dividend ratio, CF_2 is the slope from regressing $\sum_{j=1}^2 \rho^{j-1} E_t^*[\Delta d_{t+j}]$ on pd_t , and DR_{10} is the slope from regressing $-\sum_{j=1}^{10} \rho^{j-1} E_t^*[r_{t+j}]$ on pd_t . The price-earnings ratio decomposition replaces dividend growth with earnings growth expectations. Unlike the one-year decomposition, $CF_2 + DR_{10}$ does not need to be equal to one, since cash flow expectations beyond year two and return expectations beyond year ten are not observed and therefore enter neither component. Newey-West standard errors with small-sample correction are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5 shows that the pattern documented at the one-year horizon carries over to longer horizons. For the price-dividend ratio, extending the cash flow horizon from one to two years raises CF_2 from 0.196 to 0.316 in the extended sample. Ten-year return expectations produce a negative discount rate component of $DR_{10} = -0.128$. The sum $CF_2 + DR_{10} = 0.188$ indicates that a substantial fraction of price-dividend ratio variation remains attributable to expectations beyond the two-year dividend-growth and ten-year return horizons covered here. In the replication subsample, CF_2 rises to 0.661 with $DR_{10} = -0.147$, matching De La O and Myers (2021), who report $CF_2 = 0.65$ and $DR_{10} = -0.07$. One- and two-year dividend growth expectations thus account for the majority of price-dividend ratio variation in the replication subsample, while ten-year return expectations remain negative and small relative to the cash flow component.

For the price-earnings ratio, extending to two-year earnings expectations leaves the extended sample results largely unchanged: CF_2 rises from 0.775 to 0.841 and DR_{10} shifts from -0.004 to -0.061 . In the replication subsample, $CF_2 = 1.012$ and $DR_{10} = -0.080$, aligning with the original values of 0.98 and -0.08 . The sum $CF_2 + DR_{10} = 0.932$ implies that one- and two-year earnings growth expectations and ten-year return expectations together account for nearly all price-earnings ratio variation in that period. Across both price ratios and both sample windows, the extended decomposition points to the same qualitative conclusion as the one-year results:

Observed cash flow expectations appear to play a larger role than observed return expectations.

4.3 Full-Horizon Decomposition

After estimating the finite-horizon decompositions, the analysis follows De La O and Myers (2021) and extends it to a full-horizon decomposition of price ratios into cash flow news and discount rate news. As shown in equation (4), the full-horizon cash flow news is $CF = \frac{\text{cov}\left(\sum_{j=1}^{\infty} \rho^{j-1} E_t^*[\Delta d_{t+j}], pd_t\right)}{\text{var}(pd_t)}$ and the full-horizon discount rate news is $DR = \frac{\text{cov}\left(-\sum_{j=1}^{\infty} \rho^{j-1} E_t^*[r_{t+j}], pd_t\right)}{\text{var}(pd_t)}$. Because survey expectations are only observed at finite horizons, the infinite sums for CF and DR cannot be computed directly. To approximate long-horizon subjective expectations, this paper adopts the geometric decay structure of De La O and Myers (2021). For dividend growth expectations:

$$E_t^*[\Delta d_{t+1+j}] - \mu_d = \phi_d^j (E_t^*[\Delta d_{t+1}] - \mu_d) + \varepsilon_{t,j}^d, \quad (15)$$

and for return expectations:

$$E_t^*[r_{t+1+j}] - \mu_r = \phi_r^j (E_t^*[r_{t+1}] - \mu_r) + \varepsilon_{t,j}^r, \quad (16)$$

where μ_d and μ_r are the long-run mean values of dividend growth and return expectations, and ϕ_d as well as ϕ_r govern how quickly deviations from these means revert. This specification is consistent with standard asset pricing models in which fundamentals follow AR(1) processes (De La O & Myers, 2021).

Under this decay structure, the infinite sums collapse and the full-horizon decomposition reduces to a scaled version of the one-year coefficients:

$$CF = \frac{CF_1}{1 - \rho\phi_d}, \quad DR = \frac{DR_1}{1 - \rho\phi_r}, \quad (17)$$

where CF_1 and DR_1 are the one-year cash flow and discount rate components from equations (10) and (12). These expressions are valid as long as the residual terms $\sum_{j=1}^{\infty} \rho^{j-1} \varepsilon_{t,j}^d$ and $\sum_{j=1}^{\infty} \rho^{j-1} \varepsilon_{t,j}^r$ are uncorrelated with the current price-dividend ratio, a condition De La O and Myers (2021) verify using two-year dividend growth expectations and ten-year return expectations.

The persistence of dividend growth expectations, ϕ_d , is identified from the relationship between one-year and two-year subjective dividend growth expectations from I/B/E/S:

$$E_t^*[\Delta d_{t+2}] - \mu_d = \phi_d (E_t^*[\Delta d_{t+1}] - \mu_d) + v_t^d. \quad (18)$$

The persistence of return expectations, ϕ_r , is identified from the Graham-Harvey survey by exploiting the gap between one-year and ten-year return expectations. The implied expectation for years 2 through 10, $E_t^*[r_{t+2,t+10}]$, equals the difference between the ten-year and one-year

forecasts and satisfies:

$$E_t^*[r_{t+2,t+10}] - 9\mu_r = \phi_r \frac{1 - \phi_r^9}{1 - \phi_r} (E_t^*[r_{t+1}] - \mu_r) + v_t^r. \quad (19)$$

Combining equations (17)–(19) with the restriction $CF + DR = 1$, three estimation strategies are available. First, a direct CF estimation: ϕ_d is estimated from equation (18), CF is computed via equation (17), and $DR = 1 - CF$ is recovered. Second, a direct DR estimation: ϕ_r is estimated from equation (19), DR is computed via equation (17), and $CF = 1 - DR$ is recovered. Third, a joint estimation: ϕ_d and ϕ_r are jointly estimated by maximum likelihood (MLE), imposing the restriction $CF + DR = 1$. Using both cash flow and return expectations yields two independent channels for measuring CF and DR , which allows the analysis to cross-validate results across different data sources and estimation approaches.

The same full-horizon decomposition approach is applied to the log price-earnings ratio pe_t . The full-horizon cash flow news component is defined as the share of variation in pe_t that covaries with the discounted stream of expected future earnings growth, $CF = \frac{\text{cov}(\sum_{j=1}^{\infty} \rho^{j-1} E_t^*[\Delta e_{t+j}], pe_t)}{\text{var}(pe_t)}$, while the discount rate news component is defined analogously using the discounted stream of expected future returns, $DR = \frac{\text{cov}(-\sum_{j=1}^{\infty} \rho^{j-1} E_t^*[r_{t+j}], pe_t)}{\text{var}(pe_t)}$, as shown in equation (7). Following the same decay structure as for dividends, with persistence parameter ϕ_e for earnings growth expectations, the full-horizon decomposition again reduces to a scaled version of the one-year components.

$$CF = \frac{CF_1}{1 - \rho\phi_e}, \quad DR = \frac{DR_1}{1 - \rho\phi_r}, \quad (20)$$

where CF_1 and DR_1 denote the one-year cash flow and discount rate components of the price-earnings ratio variance decomposition. The persistence parameter ϕ_e is identified from the relationship between one- and two-year earnings growth expectations:

$$E_t^*[\Delta e_{t+2}] - \mu_e = \phi_e (E_t^*[\Delta e_{t+1}] - \mu_e) + v_t^e, \quad (21)$$

and ϕ_r is estimated as before from the Graham-Harvey survey. The three estimation strategies – direct CF estimation, direct DR estimation, and joint maximum likelihood – carry over exactly as described for the price-dividend ratio. De La O and Myers (2021) demonstrate that using the exact decomposition and explicitly accounting for expected future payout ratios yields virtually the same results as this approximation.

Table 6 presents the full-horizon decomposition results for both the price-dividend and price-earnings ratios under the geometric decay assumption. The full-horizon estimates for the price-dividend ratio indicate a larger role for cash flow news than for discount rate news across all estimation methods.

Table 6: Full-Horizon Variance Decomposition of S&P 500 Price Ratios

	ϕ_d / ϕ_e (1)	ϕ_r (2)	CF (3)	DR (4)
<i>Panel A: Price-Dividend Ratio</i>				
<i>Extended Sample (2003Q1–2023Q3)</i>				
Dividend growth	0.646 (0.130)		0.532* (0.285)	0.468 (0.285)
Returns		0.578 (0.158)	1.086*** (0.041)	−0.086** (0.041)
Dividend growth and returns	0.838 (0.130)	0.574 (0.158)	1.085 (0.880)	−0.085** (0.040)
<i>Replication Subsample (2003Q1–2015Q3)</i>				
Dividend growth	0.646 (0.114)		1.070*** (0.356)	−0.070 (0.356)
Returns		0.609 (0.185)	1.139*** (0.073)	−0.139* (0.073)
Dividend growth and returns	0.669 (0.114)	0.607 (0.185)	1.139*** (0.398)	−0.139* (0.073)
<i>Panel B: Price-Earnings Ratio</i>				
<i>Extended Sample (2003Q1–2023Q3)</i>				
Earnings growth	0.090 (0.021)		0.850*** (0.216)	0.150 (0.216)
Returns		0.578 (0.158)	1.008*** (0.016)	−0.008 (0.016)
Earnings growth and returns	0.236 (0.021)	0.527 (0.158)	1.007*** (0.257)	−0.007 (0.014)
<i>Replication Subsample (2003Q1–2015Q3)</i>				
Earnings growth	0.093 (0.010)		1.028*** (0.136)	−0.028 (0.136)
Returns		0.609 (0.185)	1.009*** (0.023)	−0.009 (0.023)
Earnings growth and returns	0.075 (0.010)	0.640 (0.185)	1.010*** (0.133)	−0.010 (0.025)

Note: This table reports the full-horizon variance decomposition of the price-dividend and price-earnings ratios into a cash flow component CF and a discount rate component DR , extending the finite-horizon results by extrapolating subjective expectations to all future horizons via a geometric decay structure. Three estimation approaches are reported for each ratio. The cash flow rows use exclusively the one- and two-year I/B/E/S subjective cash flow growth expectations to estimate the persistence ϕ_d (dividends) or ϕ_e (earnings), from which $CF = CF_1 / (1 - \rho\phi_d)$ or $CF = CF_1 / (1 - \rho\phi_e)$, and $DR = 1 - CF$. The returns rows use exclusively the one- and ten-year Graham-Harvey return expectations to estimate ϕ_r , from which $DR = DR_1 / (1 - \rho\phi_r)$, and $CF = 1 - DR$. The joint rows combine both sources in a maximum likelihood estimation constrained by $CF + DR = 1$, jointly identifying ϕ_d or ϕ_e and ϕ_r . In all rows, CF_1 and DR_1 are obtained by regressing one-year cash flow growth and return expectations on the current log price ratio. Standard errors for the persistence parameters in the separate rows are small-sample adjusted Newey-West; in the joint row the persistence parameters are estimated via MLE but their standard errors retain the Newey-West estimates from the separate regressions. Standard errors for CF and DR are derived from those of CF_1 , DR_1 , and the relevant persistence parameter, assuming independence. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Method 1, which estimates the cash flow share directly from dividend growth expectations, yields $CF = 0.532$ in the extended sample, significant at the 10% level. This estimate is in line with the progression observed in the finite-horizon decompositions ($CF_1 = 0.196$, $CF_2 = 0.316$; both significant at the 5% level), indicating that longer horizons assign a greater role to cash flow expectations. In the replication subsample, Method 1 yields $CF = 1.070$, comparable to the estimate of De La O and Myers (2021) of 0.93. The substantially higher estimate reflects the tighter link between dividend growth expectations and the price-dividend ratio in that period. As documented in the one-year horizon decomposition, the co-movement coefficient increases from 0.196 in the extended sample to 0.394 in the replication subsample, so the increase in CF_1 directly translates into a greater full-horizon CF via equation (17). Since the persistence parameter $\phi_d = 0.646$ is stable across both samples – consistent with De La O and Myers (2021) ($\phi_d = 0.59$) – the scaling factor $1/(1 - \rho\phi_d)$ is identical in both samples. This implies that the difference in full-horizon CF between the two samples is driven entirely by the increase in CF_1 .

Notably, the CF estimate under Method 1 in the extended sample (0.532) is lower than the values obtained under Methods 2 and 3, which produce cash flow shares of $CF = 1.086$ and $CF = 1.085$, with discount rate components close to zero and negative ($DR = -0.086$ and $DR = -0.085$). Under Method 2, $DR_1 < 0$ and subjective return expectations have low volatility, hence DR is estimated to be close to zero and negative. The identity $CF = 1 - DR$ then mechanically forces the cash flow share above one. Under Method 3, the joint MLE is similarly constrained: DR_1 is close to zero and therefore no adjustment of ϕ_r can substantially shift DR away from its near-zero negative value. This leaves CF above one as well. The nearly identical results of Methods 2 and 3 follow directly from the magnitude of DR in both cases.

In the replication subsample of Panel A, all three methods converge to CF between 1.070 and 1.139, all significant at the 1% level, comparable to those reported by De La O and Myers (2021). For the extended sample, the results qualify their conclusion: Cash flow news remains the dominant driver and the discount rate channel negligible. However, the tight link between short-term dividend growth expectations and the price-dividend ratio weakens considerably after 2015. Figure 2 illustrates this directly. Cash flow expectations track several cyclical movements in the price-dividend ratio, most notably the sharp contraction and recovery around 2008–2009 during the global financial crisis. Yet a clear divergence emerges from the mid-2010s onward. Especially after 2015, the price-dividend ratio trends upward while dividend growth expectations remain subdued. This decoupling explains why Method 1 accounts for only about half of the price-dividend ratio variation in the extended sample, with the remainder reflecting price ratio movements not captured by short-term survey expectations alone.

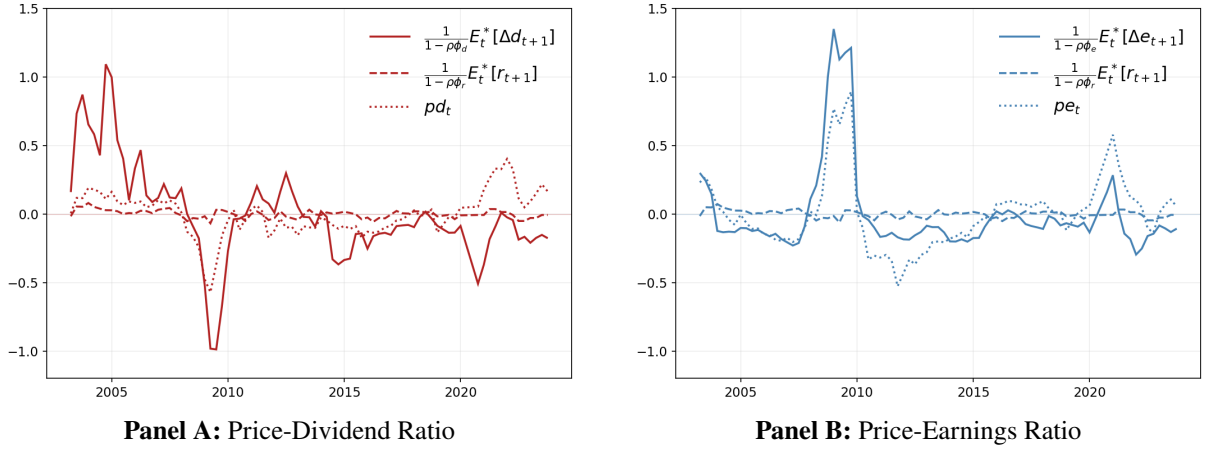


Figure 2: Price Ratios and Full-Horizon Subjective Expectations

Note: This figure compares the log price ratios to their corresponding full-horizon subjective expectations, all constructed from survey data. Panel A shows the log price-dividend ratio pd_t alongside the full-horizon dividend growth expectation $\frac{1}{1-\rho\phi_d}E_t^*[\Delta d_{t+1}]$ and the full-horizon return expectation $\frac{1}{1-\rho\phi_r}E_t^*[r_{t+1}]$. Panel B shows the analogous series for the price-earnings ratio, using earnings growth expectations $\frac{1}{1-\rho\phi_e}E_t^*[\Delta e_{t+1}]$. All variables are demeaned. Persistence parameters ϕ_d , ϕ_e , and ϕ_r are taken from the joint MLE estimation (Method 3). The solid line plots cash flow expectations, the dashed line plots return expectations, and the dotted line plots the observed price ratio. The sample runs from 2003Q1 to 2023Q3.

Panel B of Figure 2 shows the analogous series for the price-earnings ratio. In contrast to Panel A, earnings growth expectations track the price-earnings ratio more closely throughout the sample. The correspondence also weakens after 2015, but the overall co-movement remains tighter than for dividends. This finding anticipates the higher and more stable cash flow shares of the variance decomposition documented across all three methods below.

The full-horizon decomposition of the price-earnings ratio again identifies cash flow news as the dominant driver. Under Method 1, the cash flow share in the extended sample is estimated at $CF = 0.850$, significant at the 1% level. While this represents a slight increase relative to the finite-horizon estimates ($CF_1 = 0.775$, $CF_2 = 0.841$), it deviates from the replication subsample value of $CF = 1.028$. As documented in the one-year horizon decomposition, this deviation reflects a contraction in the volatility of earnings growth expectations and a weakening of their correlation with the price-earnings ratio after 2015. Both effects jointly depress CF_1 in the extended sample. Moreover, since ϕ_e is stable across both samples ($\phi_e = 0.090$ in the extended sample vs. $\phi_e = 0.093$ in the replication subsample), the scaling factor $1/(1 - \rho\phi_e)$ is nearly unchanged, and the lower full-horizon CF under Method 1 is therefore attributable to the lower CF_1 .

Methods 2 and 3 produce higher cash flow shares of $CF = 1.008$ and $CF = 1.007$, with discount rate components close to zero and negative ($DR = -0.008$ and $DR = -0.007$). The same mechanism as for the price-dividend ratio applies: DR_1 is near zero, which constrains DR to remain small relative to the cash flow component and mechanically forces CF above one via the identity $CF + DR = 1$. However, the gap between Method 1 and Methods 2 and 3 is smaller for

the price-earnings ratio.

A noteworthy property of the earnings decomposition relative to dividends concerns persistence. The estimated persistence of earnings growth expectations is substantially lower across both samples ($\phi_e = 0.090$ in the extended sample and $\phi_e = 0.093$ in the replication subsample, compared with $\phi_d = 0.646$ in both samples), matching the estimate of $\phi_e = 0.06$ reported by De La O and Myers (2021). This reflects the phenomenon of dividend smoothing: Firms adjust dividends gradually instead of immediately passing through earnings shocks, hence revisions to expected dividend growth persist across horizons, whereas revisions to expected earnings growth are perceived as largely transitory. The low persistence of earnings growth expectations also explains why the incremental gain from extending the horizon is small for earnings: Because ϕ_e is close to zero, the scaling factor $1/(1 - \rho\phi_e)$ approaches one. That means that almost all of the full-horizon cash flow share is already captured at the one-year horizon.

In the replication subsample, all three methods converge to cash flow shares close to one: $CF = 1.028$ under Method 1 and $CF = 1.009$ and $CF = 1.010$ under Methods 2 and 3, all significant at the 1% level. They match the original estimates of 0.99, 1.01, and 1.01 of De La O and Myers (2021). Overall, price-earnings ratio fluctuations are predominantly driven by revisions in subjective earnings growth expectations rather than by changes in discount rate expectations, a conclusion that holds robustly across all three methods and both sample periods.

4.4 Earnings Noise, Prices, and Long-Term Expectations

The results confirm the central finding of De La O and Myers (2021) that cash flow expectations, instead of discount rate expectations, are the dominant source of price ratio fluctuations. Focusing on earnings, one notable result is that in the replication subsample one-year earnings growth expectations account for nearly all variation in the price-earnings ratio ($CF_1 = 0.935$) and the discount rate contribution is close to zero. Extending to the full sample reveals a weaker picture: The discount rate channel remains small relative to the cash flow component, but the contribution of short-term cash flow expectations declines ($CF_1 = 0.775$), driven in part by a contraction in earnings growth volatility from 31.1% to 25.1% once the post-2015 period is included.

The finding of a lower CF_1 in the extended sample connects to a broader methodological point raised by Nagel (2024): Variance decompositions of price ratios explain movements in the ratio itself, not directly in the stock price level. For the price-earnings ratio in particular, a substantial part of ratio variation can originate from fluctuations in the earnings denominator rather than in prices. Short-term earnings expectations are strongly mean-reverting ($\phi_e \approx 0.09$) and are closely linked to transitory earnings components – such as special items that become strongly negative in recessions – which mechanically drive up the price-earnings ratio while analysts simultaneously forecast a sharp earnings recovery. This tends to produce a strong positive co-movement between short-term earnings expectations and the price-earnings ratio that is largely mechanical. Under this interpretation, a high variance share of short-term expectations reflects their co-movement

with the earnings denominator rather than with the price numerator, and thus captures earnings dynamics instead of price movements (Adam & Nagel, 2022; Nagel, 2024).

Nagel (2024) provides supporting evidence by showing that once special items are excluded from the earnings definition, the variance share attributed to short-term growth expectations declines from around 52% to 16%, suggesting that much of the high variance share documented by De La O and Myers (2021) is driven by quickly reverting earnings noise instead of price movements associated with short-term growth expectations. The lower CF_1 in the extended sample is compatible with this interpretation, as the post-2015 period lacks the severe earnings dislocations that are present in the replication subsample and associated with the high cash flow share documented there.

This consideration raises the question of what actually drives aggregate stock prices. Bordalo et al. (2020, 2024) argue that long-term instead of short-term growth expectations are the primary source of persistent price movements and predictable returns, precisely because they are not dominated by transitory earnings dynamics. The apparent contradiction between De La O and Myers (2021) and Bordalo et al. (2020) dissolves once one recognizes, according to Adam and Nagel (2022), that variance decompositions of the price-earnings ratio capture movements in the earnings denominator as much as movements in prices. Nevertheless, the literature agrees that cash flow expectations dominate discount rate expectations in driving asset prices (Adam & Nagel, 2022). What remains unclear is whether short- or long-term cash flow components drive stock prices – a question that cannot be answered by studying price ratio movements alone.

Bordalo et al. (2024) state that return predictability provides a way to distinguish between the two. The variance decomposition has established that short-term cash flow expectations dominate price ratio variation, but cannot determine whether this reflects tracking of transitory earnings dynamics or systematic belief distortions that temporarily inflate prices. If short-term expectations merely track current earnings, prices would reflect fundamentals and future returns would be unpredictable. If instead long-term expectations reflect systematic overreaction to news – as argued by Bordalo et al. (2024) – they temporarily inflate prices above fundamental value, and the eventual correction should generate predictably low future returns. Return predictability thus reveals which horizon of cash flow expectations drives stock prices and whether this reflects information or mispricing (Bordalo et al., 2024).

The following analysis focuses exclusively on earnings expectations and the price-earnings ratio for two reasons. The contrast between short- and long-term expectation components is most pronounced for earnings compared to dividends, because one-year earnings growth expectations account for nearly all price-earnings ratio variation. Furthermore, the return predictability literature on which the analysis draws (e.g. Adam & Nagel, 2022; Bordalo et al., 2024) focuses primarily on earnings expectations, ensuring direct comparability with existing evidence.

5. Cash Flow Expectations and Return Predictability

In this section, the paper tests if short- and long-term cash flow expectations exhibit return predictability. The goal is to determine whether these cash flow expectations reflect information about fundamentals or systematic belief distortions that cause prices to diverge from their fundamental value.

5.1 Return Predictability Across Forecast Horizons

Drawing on the approach of Bordalo et al. (2024), future cumulative aggregate stock returns over one-, three-, and five-year horizons are regressed on three earnings growth expectation measures: one-year, two-year, and long-term earnings growth expectations. If short-term expectations primarily track transitory earnings dynamics, they should exhibit little predictive power for future returns. If, by contrast, long-term expectations reflect systematic overreaction to recent news, their distortions should be corrected over time and thus forecast future returns (Bordalo et al., 2024). To ensure comparability with both strands of the literature, two sample windows are used: the Bordalo et al. (2024) sample running from 1981Q1 to 2015Q4, and the replication subsample of De La O and Myers (2021) from 2003Q1 to 2015Q3. Table 7 reports the results.

Panel A shows a negative and statistically significant relationship between long-term growth expectations and future returns across both sample windows. In the 1981–2015 window, LTG is negatively related to future returns at all horizons, with coefficients of -0.321 , -0.502 , and -0.514 for one-, three-, and five-year returns, significant at the 5% and 1% levels respectively, and explaining 26% of five-year return variation. This is consistent with Bordalo et al. (2024), who show that long-term growth expectations strongly predict future returns while short-term expectations exhibit little predictive power. They thereby argue that long-term expectations reflect belief-driven valuation cycles. The estimated coefficients are slightly larger in magnitude than those of Bordalo et al. (2024) (-0.239 , -0.402 , and -0.435), likely reflecting differences in data frequency and aggregation methodology.⁵ In the replication subsample, the negative relationship persists and strengthens at longer horizons: The three- and five-year coefficients are -0.497 and -0.698 , significant at the 5% and 1% levels respectively, with the five-year adjusted R^2 rising to 48%. The one-year coefficient is not statistically significant in this shorter window.

Panels B and C show a different pattern for short-term expectations. In the 1981–2015 window, one-year earnings growth expectations exhibit no statistically significant predictive power for returns at any horizon, consistent with adjusted R^2 values close to zero. In the replication subsample, one-year expectations again show no significant predictive power at the one- and three-

⁵Bordalo et al. (2024) use monthly instead of quarterly data and aggregate firm-level EPS forecasts by multiplying each forecast by shares outstanding, summing across all S&P 500 firms, and dividing by the total index share count. Beyond the second fiscal year, they assume earnings grow at the firm-level LTG rate. This paper instead scales covered-firm aggregates by the market-capitalization ratio and applies linear interpolation between adjacent fiscal-year forecasts without LTG extrapolation beyond year two.

year horizons, though the five-year coefficient (0.395, adj. $R^2 = 14\%$) reaches the 5% significance level. This result should be interpreted with caution, because the replication subsample coincides with earnings dislocations around the financial crisis, where transitory earnings collapses and recoveries may drive the apparent predictability instead of belief distortions. Two-year earnings growth expectations display slightly more predictive power at longer horizons in both samples: The five-year coefficients are significant at the 1% level (0.489 in the 1981–2015 window and 0.528 in the replication subsample), with adjusted R^2 values of 23% and 26%, respectively. This pattern broadly differs from Bordalo et al. (2024), who find only a single significant coefficient across all short-term specifications. They observe a significant positive effect of one-year earnings expectations on five-year returns with an adjusted R^2 of 3%, underscoring that the short-term channel remains economically negligible in their sample.

Table 7: Earnings Growth Expectations and Aggregate Stock Return Predictability

	Sample: 1981Q1–2015Q4			Sample: 2003Q1–2015Q3		
	r_{t+1}	$\sum_{j=1}^3 \rho^{j-1} r_{t+j}$	$\sum_{j=1}^5 \rho^{j-1} r_{t+j}$	r_{t+1}	$\sum_{j=1}^3 \rho^{j-1} r_{t+j}$	$\sum_{j=1}^5 \rho^{j-1} r_{t+j}$
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Returns and LTG</i>						
LTG _t	−0.321** (0.123)	−0.502*** (0.112)	−0.514*** (0.096)	−0.436 (0.271)	−0.497** (0.214)	−0.698*** (0.103)
Adj. R^2 (%)	10	25	26	17	23	48
<i>Panel B: Returns and Growth Forecast for Year 1</i>						
$\mathbb{E}_t^*[\Delta e_{t+1}]$	−0.193 (0.170)	−0.016 (0.103)	0.137 (0.096)	0.066 (0.130)	0.199 (0.149)	0.395** (0.172)
Adj. R^2 (%)	3	−1	1	−2	2	14
<i>Panel C: Returns and Growth Forecast for Year 2</i>						
$\mathbb{E}_t^*[\Delta e_{t+2}]$	0.122 (0.105)	0.282** (0.115)	0.489*** (0.095)	0.023 (0.143)	0.307* (0.162)	0.528*** (0.120)
Adj. R^2 (%)	1	7	23	−2	8	26

Note: Each panel reports slope coefficients from univariate regressions of future S&P 500 returns on standardized earnings growth expectations, estimated separately over two sample windows. The dependent variable in columns (1) and (4) is the log one-year return r_{t+1} ; in the remaining columns it is the discounted cumulative return $\sum_{j=1}^h \rho^{j-1} r_{t+j}$ for $h \in \{3, 5\}$ years, where the discount factor $\rho = e^{\overline{p\overline{d}}}/(1 + e^{\overline{p\overline{d}}})$ is computed from the sample mean log price-dividend ratio as described in Section 2 ($\rho = 0.977$). Panel A uses the aggregate LTG forecast as predictor; Panels B and C use one- and two-year-ahead earnings growth forecasts. All predictors are standardized to mean zero and unit standard deviation within each sample window, intercepts are not reported. Columns (1)–(3) cover the Bordalo et al. (2024) sample (1981Q1–2015Q4); columns (4)–(6) cover the replication subsample of De La O and Myers (2021) (2003Q1–2015Q3). Small-sample adjusted Newey-West standard errors are reported in parentheses, with lags of 4, 12, and 20 quarters for the one-, three-, and five-year return horizons respectively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Overall, these results support the findings of Bordalo et al. (2024) that one-year earnings growth expectations do not robustly predict future returns. The absence of predictive power, with the exception of the five-year horizon in the replication subsample, is consistent with the view that

the high co-movement between one-year earnings expectations and the price-earnings ratio primarily reflects transitory earnings dynamics. Long-term growth expectations, on the other hand, negatively predict future returns across both samples and all horizons, with the exception of the one-year horizon in the replication subsample.

The results of this and the preceding sections carry an implicit assumption that long-term growth expectations – measured by the I/B/E/S long-term growth forecast LTG – capture investors’ subjective beliefs. Nagel (2024) raises the concern that analysts may construct their LTG estimates not by forming independent views about long-run fundamentals but by inferring the growth rate implied by the current stock price while erroneously treating the discount rate as fixed. In this case, LTG would mechanically co-move with discount rates rather than with subjective cash flow beliefs and reflect time-varying required returns. This concern is addressed by testing whether LTG constitutes a proxy for investors’ subjective expectations and whether its predictive power reflects nonrational belief distortions instead of rationally time-varying required returns.

5.2 Robustness: Controlling for Time-Varying Required Returns

If LTG reflects time-varying required returns and not belief distortions, its predictive power for future returns should be absorbed once proxies for required returns are included as controls (Bordalo et al., 2024). The baseline LTG regressions of Table 7 are therefore augmented with the log price-dividend ratio and the log price-earnings ratio as such controls, based on Bordalo et al. (2024). If the LTG coefficient remains significant after their inclusion, this would indicate that LTG carries independent information about future returns beyond what is already reflected in current valuations. This finding would suggest that LTG captures a belief component that is not fully reflected in the price ratios.

Table 8 reports the results for the five-year return horizon, at which LTG displays the strongest predictive power in Table 7. The evidence broadly supports the interpretation of Bordalo et al. (2024) that LTG captures belief distortions, though the picture differs across the two price ratio controls. In Panel A, the LTG coefficient remains negative in both samples after controlling for the log price-dividend ratio, but reaches statistical significance only in the replication subsample (-0.390 , significant at the 1% level). The coefficient in the 1981–2015 window (-0.197) does not reach conventional significance levels. This suggests that in the longer sample LTG and the price-dividend ratio capture overlapping information, which contrasts with the results of Bordalo et al. (2024), who find a significant LTG coefficient of -0.235 even after controlling for the price-dividend ratio. Bordalo et al. (2024) point out, however, that this is a demanding test: Stock prices already incorporate market expectations, so the price-dividend ratio captures much of the same information as LTG, and controlling for market prices may absorb its predictive power even if beliefs are the true driving force.

The same information overlap is evident from the opposite direction: When LTG is introduced alongside the price-dividend ratio, it extracts the shared belief component and causes the ratio’s

coefficient to decline from -0.703 to -0.599 in the 1981–2015 window and from -0.763 to -0.540 in the replication subsample. The residual predictive power of the price-dividend ratio after controlling for LTG likely reflects variation in required returns orthogonal to the belief component captured by LTG (Bordalo et al., 2024).

Table 8: Return Predictability: Price Ratios and LTG

	Sample: 1981Q1–2015Q4		Sample: 2003Q1–2015Q3	
	(1)	(2)	(3)	(4)
<i>Panel A: $\log(P/D)$ as control</i>				
LTG_t		-0.197 (0.124)		-0.390^{***} (0.128)
$\log(P/D)_t$	-0.703^{***} (0.135)	-0.599^{***} (0.149)	-0.763^{***} (0.138)	-0.540^{***} (0.168)
Adj. R^2 (%)	49	52	57	67
<i>Panel B: $\log(P/E)$ as control</i>				
LTG_t		-0.441^{***} (0.122)		-0.709^{***} (0.085)
$\log(P/E)_t$	-0.363^{**} (0.176)	-0.206 (0.157)	0.145 (0.235)	-0.042 (0.164)
Adj. R^2 (%)	13	29	0	47

Note: Each column reports slope coefficients from bivariate regressions of discounted five-year cumulative returns $\sum_{j=1}^5 \rho^{j-1} r_{t+j}$ on LTG and a price ratio, estimated separately over two sample windows. The discount factor $\rho = 0.977$ is derived from the sample mean log price-dividend ratio as described in Section 2. Columns (1) and (3) report the univariate benchmark regressions of five-year returns on the price ratio alone; columns (2) and (4) augment these with LTG_t as an additional predictor. Panel A uses the log price-dividend ratio as the valuation control; Panel B uses the log price-earnings ratio. All predictors are standardized to mean zero and unit standard deviation within each sample. The adjusted R^2 reported for the univariate columns serves as a benchmark against which the incremental explanatory power of LTG can be assessed. Small-sample adjusted Newey-West standard errors with a lag of 20 quarters are reported in parentheses. $N = 137$ for the Bordalo et al. (2024) sample of 1981Q1–2015Q4 and $N = 51$ for the replication subsample of De La O and Myers (2021). $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Panel B considers the price-earnings ratio. In the 1981–2015 window, the ratio is significant when considered alone (coef. = -0.363). Once LTG is included, it becomes statistically insignificant and LTG itself is significant at the 1% level (coef. = -0.441). These results mirror those of Bordalo et al. (2024), who report an LTG coefficient of -0.468 and a similarly insignificant price-earnings ratio coefficient of -0.180 under the same control. In the replication subsample, the price-earnings ratio yields an insignificant coefficient of 0.145 when considered alone. LTG, in contrast, remains strongly significant at -0.709 . This indicates that in the period examined by De La O and Myers (2021), return predictability stems from the belief component captured in LTG rather than from the price ratio itself. Overall, LTG retains its predictive power in both samples after including the price-earnings ratio.

Altogether, the results support the perspective of Bordalo et al. (2024) that LTG captures systematic belief distortions instead of time-varying required returns. LTG retains independent predictive power after controlling for price ratios, though not for the price-dividend ratio in the

1981–2015 window, and absorbs part of the explanatory power of price ratios. Consequently, the question emerges how these belief distortions arise. The negative return predictability documented in this Section 5 suggests that long-term growth expectations are systematically too optimistic: Periods of elevated optimism are followed by disappointing returns as inflated beliefs correct toward fundamentals, according to the overreaction mechanism described in Bordalo et al. (2019, 2024). The next section therefore applies the Coibion-Gorodnichenko framework to test directly whether analyst forecasts systematically over- or underreact at the level of the belief formation process.

6. Forecast Rationality and Belief Distortions

6.1 Coibion-Gorodnichenko Regression

The Coibion-Gorodnichenko framework (Coibion & Gorodnichenko, 2012, 2015) provides a direct test of forecast rationality by examining whether forecast errors are predictable from forecast revisions. Based on Bordalo et al. (2019, 2020), the paper applies this framework to long-term growth expectations to assess whether the return predictability evidence of Section 5 reflects overreaction in the expectation formation process. The forecast error is defined as the difference between realized annualized log earnings growth over n years and the long-term growth expectation held at time t , following Bordalo et al. (2020):

$$FE_{t,n}^{\text{LTG}} = \frac{1}{n}(\log E_{t+n} - \log E_t) - \text{LTG}_t, \quad (22)$$

where E_t denotes the S&P 500 earnings index at time t , constructed following Section 2, and $n \in \{3, 4, 5\}$ years. The forecast revision over $k \in \{1, 2, 3\}$ years is:

$$\Delta_k \text{LTG}_t = \text{LTG}_t - \text{LTG}_{t-k}, \quad (23)$$

yielding the regression:

$$FE_{t,n}^{\text{LTG}} = \alpha + \beta_k \Delta_k \text{LTG}_t + \varepsilon_{t+n}. \quad (24)$$

Under full-information rational expectations, $\beta_k = 0$: Past revisions carry no predictive power for subsequent forecast errors. A negative coefficient $\beta_k < 0$ indicates overreaction: Upward revisions to LTG are followed by realized earnings growth that falls systematically short of expectations. A positive coefficient $\beta_k > 0$ indicates underreaction: Upward revisions to LTG are followed by realized earnings growth that systematically exceeds expectations, as forecasters incorporate new information too slowly.

Table 9 shows the estimated $\hat{\beta}_k$ coefficients across all combinations of forecast horizon $n \in \{3, 4, 5\}$ years and revision lag $k \in \{1, 2, 3\}$ years, estimated over the full available LTG sample. The results support the presence of overreaction in aggregate long-term earnings growth expectations,

but the evidence weakens as the revision lag increases. All nine estimated coefficients are negative. For one-year revisions (Panel A), $\hat{\beta}_1$ ranges from -5.457 at the three-year horizon to -3.582 at the five-year horizon. All three coefficients are statistically significant at the 5% level. For two-year revisions (Panel B), $\hat{\beta}_2$ ranges from -4.241 to -1.868 ; the three- and four-year horizon estimates are significant at the 1% and 5% levels respectively and the five-year coefficient is significant at the 10% level. For three-year revisions (Panel C), only the three-year horizon estimate ($\hat{\beta}_3 = -3.122$) remains significant at the 1% level and the four-year estimate at the 10% level ($\hat{\beta}_3 = -1.650$). The five-year coefficient is no longer statistically different from zero ($\hat{\beta}_3 = -0.485$).

Table 9: Coibion-Gorodnichenko Regressions for Earnings LTG

	Dependent Variable		
	$\frac{1}{3}(\log E_{t+3} - \log E_t) - \text{LTG}_t$ (1)	$\frac{1}{4}(\log E_{t+4} - \log E_t) - \text{LTG}_t$ (2)	$\frac{1}{5}(\log E_{t+5} - \log E_t) - \text{LTG}_t$ (3)
<i>Panel A: $\Delta_1\text{LTG}_t = \text{LTG}_t - \text{LTG}_{t-1}$</i>			
$\Delta_1\text{LTG}_t$	-5.457** (2.163)	-4.665** (1.742)	-3.582** (1.615)
<i>N</i>	152	148	144
<i>Sample</i>	1982Q4–2020Q3 / 1982Q4–2019Q3 / 1982Q4–2018Q3		
<i>Panel B: $\Delta_2\text{LTG}_t = \text{LTG}_t - \text{LTG}_{t-2}$</i>			
$\Delta_2\text{LTG}_t$	-4.241*** (1.176)	-3.024** (1.102)	-1.868* (0.962)
<i>N</i>	148	144	140
<i>Sample</i>	1983Q4–2020Q3 / 1983Q4–2019Q3 / 1983Q4–2018Q3		
<i>Panel C: $\Delta_3\text{LTG}_t = \text{LTG}_t - \text{LTG}_{t-3}$</i>			
$\Delta_3\text{LTG}_t$	-3.122*** (1.073)	-1.650* (0.987)	-0.485 (1.002)
<i>N</i>	144	140	136
<i>Sample</i>	1984Q4–2020Q3 / 1984Q4–2019Q3 / 1984Q4–2018Q3		

Note: This table reports OLS slope coefficients $\hat{\beta}_k$ from the regression $FE_{t,n}^{\text{LTG}} = \alpha + \beta_k \Delta_k \text{LTG}_t + \varepsilon_{t+n}$ across all combinations of forecast horizon $n \in \{3, 4, 5\}$ years and revision lag $k \in \{1, 2, 3\}$ years. The dependent variable is the forecast error, calculated as the annualized realized earnings growth rate over n years minus the LTG forecast held at time t , i.e. $FE_{t,n}^{\text{LTG}} = \frac{1}{n}(\log E_{t+n} - \log E_t) - \text{LTG}_t$. The independent variable is the forecast revision, calculated as the change in the aggregate LTG forecast over k years, $\Delta_k \text{LTG}_t = \text{LTG}_t - \text{LTG}_{t-k}$. Small-sample adjusted Newey-West standard errors are reported in parentheses, with lag length of 12, 16, and 20 quarters for the three-, four-, and five-year horizons respectively, based on Bordalo et al. (2020). The sample start dates reflect the construction of the revision variable: With a revision lag of k years, the first usable observation requires LTG data from k years prior to the forecast date. Since the LTG series begins in 1981Q4, Panel A ($k = 1$) starts in 1982Q4, Panel B ($k = 2$) in 1983Q4, and Panel C ($k = 3$) in 1984Q4. Correspondingly, the sample end dates are n years before the full-sample endpoint of 2023Q3, since the forecast error requires n years of realized earnings growth following each forecast date; for $n \in \{3, 4, 5\}$ this yields end dates of 2020Q3, 2019Q3, and 2018Q3, respectively. $^* p < 0.10$, $^{**} p < 0.05$, $^{***} p < 0.01$.

The sign of the Panel A estimates is consistent with Bordalo et al. (2020), who report coefficients of -10.073 , -6.818 , and -5.100 for the three-, four-, and five-year horizons using one-year

revisions, all significant at the 1% level. The scale of the estimated coefficients differs, however, likely reflecting differences in index construction.⁶ The findings also deviate from Bordalo et al. (2019), who document monotonically increasing overreaction coefficients at the firm level across both horizons n and k . They attribute this pattern to the growing difficulty of predicting earnings growth further into the long term. In the aggregate regressions, $\hat{\beta}_k$ instead becomes less negative and loses significance as the revision lag k as well as the forecast horizon n increase. This might suggest that aggregation to the index level averages out the cross-sectional heterogeneity in overreaction intensity that drives the firm-level pattern.

The negative estimates of $\hat{\beta}_k$ imply that forecasters revise their long-term growth expectations too strongly, such that upward revisions are systematically followed by earnings outcomes that fall below the revised forecasts. This finding suggests that analysts place excessive weight on recent information when updating their beliefs, causing forecasts to overshoot the rational benchmark. The results are in line with the evidence that long-horizon forecasts exhibit overreaction (Bastianello, 2025; Bordalo et al., 2019, 2024, 2025; La Porta, 1996). According to Bastianello (2025), Bordalo et al. (2024), and La Porta (1996), the forecast errors imply a negative relationship between belief bias and subsequent returns. This links the overreaction identified here directly to the negative return predictability documented in Section 5.

6.2 Discussion of Coibion-Gorodnichenko Regression Findings

A potential concern with the Coibion-Gorodnichenko test is that a negative $\hat{\beta}_k$ could arise mechanically from noise in LTG forecasts instead of overreaction. De Silva and Thesmar (2024) show that at the individual analyst level, both bias and noise increase with forecast horizon, and that this noise inflation can produce a negative coefficient at longer horizons even in the absence of any systematic over- or underreaction. At the aggregate level, this concern is mitigated. De Silva and Thesmar (2024) themselves find that idiosyncratic noise across forecasters is averaged out as the number of analysts in the consensus increases. Since the LTG index is a value-weighted average across a large cross-section of S&P 500 analysts, the individual-level noise inflation is reduced. However, any component of noise that is correlated across forecasters would survive aggregation and cannot be fully ruled out.

Finally, a more fundamental challenge remains in identifying whether the mechanism underlying the results reflects belief distortions or rational variation in risk premia. In the present-value framework, the latter corresponds to rationally time-varying required returns. The Coibion-Gorodnichenko estimates of a negative $\hat{\beta}_k$ and the negative return predictability of LTG documented in Section 5 are interpreted by Bastianello (2025), Bordalo et al. (2024), and La Porta (1996) as two outcomes of the same underlying mechanism: Excess optimism in long-term growth

⁶Bordalo et al. (2020) aggregate at monthly instead of quarterly frequency, weight firm-level forecasts by shares outstanding divided by the total index share count, and extrapolate earnings beyond the second fiscal year using the firm-level LTG rate.

expectations inflates prices beyond fundamental value, and when overly optimistic forecasts are subsequently disappointed, beliefs and prices correct downward. This mechanism generates both the negative forecast errors and the negative return predictability observed. Bordalo et al. (2024) formalize this link more directly by instrumenting future returns with the predicted forecast error from the Coibion-Gorodnichenko regression and find that periods of excess optimism are followed by significantly lower stock returns even after controlling for the price-dividend ratio. They interpret this result as evidence that the link between forecast errors and return predictability operates through belief distortions instead of rational variation in risk premia.

Nagel (2024), however, argues that this pattern of upward LTG revisions, negative forecast errors, and negative future returns is equally consistent with analysts mechanically backing out growth expectations from prices temporarily elevated by low risk premia: Falling required returns raise prices, analysts fit a higher LTG to justify them, and when risk premia normalize, prices fall and earnings disappoint relative to the inflated LTG estimate. Bordalo et al. (2024) attempt to refute this by showing that LTG retains return predictability after controlling for price ratios and that LTG revisions respond to earnings news beyond returns. Nagel (2024) counters, however, that neither test is conclusive. If risk premia covary with earnings growth, the earnings news regression is consistent with price inference, and if price ratios imperfectly proxy for risk premia, LTG will retain incremental predictive power under both mechanisms.

The two mechanisms, belief distortion and rational variation in risk premia, are therefore observationally equivalent with the observed data. Whether LTG constitutes a proxy for investors' subjective beliefs or reflects time-varying required returns embedded in market prices remains an open question. Nevertheless, Nagel (2024) concedes that several pieces of evidence are difficult to reconcile with pure price inference. First, there is substantial cross-sectional disagreement in LTG across analysts. Because all analysts observe the same market price, a price-fitting procedure would impose a tighter consensus on growth estimates – unless analysts additionally differ in their assumed discount rates. Moreover, if analysts disagree on the weighted average cost of capital (WACC), those assuming a higher discount rate should compensate with lower growth expectations to rationalize approximately the same market price. This implies a negative correlation between WACC and LTG across analysts. Décaire et al. (2024), cited in Nagel (2024), examine analysts who rely on discounted cash flow models and find no negative correlation between assumed discount rates and growth forecasts. Nagel (2024) argues that this finding is inconsistent with the price-inference hypothesis.

Overall, the results of the negative return predictability of LTG, its robustness to price ratio controls, and the Coibion-Gorodnichenko evidence of systematic overreaction are compatible with the belief-distortion mechanism of Bordalo et al. (2024). Nevertheless, the identification challenge raised by Nagel (2024) cannot be fully resolved with the available data. It remains an open question whether LTG constitutes a valid proxy for subjective expectations, rather than a reflection of time-varying required returns. As Nagel (2024) underlines, a definitive

distinction will require better proxies for subjective expectations, for instance direct survey evidence from investors instead of analysts. Beyond that, a more detailed understanding of how analysts construct their forecasts and what drives the substantial disagreement observed across them would help clarify whether reported LTG reflects beliefs about long-run fundamentals or captures time-varying required returns.

7. Conclusion

This paper examines whether analyst expectations, disaggregated by forecast horizon, explain S&P 500 price ratios and aggregate return predictability. The results are horizon-dependent: Short-term cash flow expectations account for most of the price ratio variation but seem to track transitory earnings dynamics instead of persistent price movements. First, the decline in the short-term cash flow share once the post-2015 years are included suggests that the strong link between short-term expectations and price ratios documented by De La O and Myers (2021) in the replication subsample over the period of 2003Q1–2015Q3 is sample-dependent and partly driven by transitory earnings volatility rather than persistent price movements. Second, they appear to carry little predictive power for future returns. In contrast, long-term growth expectations co-move less with price ratios yet negatively predict returns at three- and five-year horizons. This predictive power remains after controlling for the price-earnings ratio but weakens when the price-dividend ratio is included in the Bordalo et al. (2024) sample period of 1981Q1–2015Q4, which may reflect the difficulty of cleanly separating belief distortions from time-varying required returns (Bordalo et al., 2024). Third, the Coibion-Gorodnichenko regressions provide evidence of systematic overreaction in long-term expectations. Taken together, these findings are consistent with the view of Bordalo et al. (2019, 2024) that belief distortions in long-term expectations may be an important contributor to aggregate return predictability: Overreaction to good news can temporarily inflate stock prices beyond fundamental value, and the subsequent correction of overly optimistic beliefs is associated with predictably low future returns. This paper contributes to the literature by extending the variance decomposition beyond 2015, linking it directly to the return predictability and forecast rationality evidence.

One important limitation remains. Whether LTG constitutes a proxy for subjective beliefs or mechanically reflects time-varying required returns embedded in market prices cannot be resolved with the available data (Nagel, 2024). The two interpretations are observationally equivalent at the aggregate level, and a definitive distinction would require direct survey evidence from investors or a more detailed understanding of how analysts construct their long-run forecasts. Future research that links individual analyst forecasting behavior to market outcomes, or that exploits cross-sectional variation in analyst disagreement as an instrument for belief dispersion, may help resolve this identification challenge and further clarify the role of expectation formation in aggregate asset pricing.

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Appendix

Appendix 1: Aggregation Methodology

This section provides the implementation details for constructing realized and expectations indices for dividends and earnings of the S&P 500, based on the methodology of De La O and Myers (2021). The construction starts with the realized dividend index before turning to the dividend expectations measure. The earnings indices are built analogously.

For each period t , the following firm-level variables are used: ordinary dividends per share $D_{i,t}$, share price $P_{i,t}$, and shares outstanding $S_{i,t}$ for each constituent company i , where $\mathcal{X}_t$ denotes the set of S&P 500 index members at time t . Total index market capitalization and aggregate dividends are computed by summing across all constituents at quarterly frequency:

$$M_t = \sum_{i \in \mathcal{X}_t} P_{i,t} S_{i,t}, \quad D_t = \sum_{i \in \mathcal{X}_t} D_{i,t} S_{i,t}. \quad (\text{A1})$$

The index level $SP500_t$ is constructed by scaling total market capitalization M_t of all S&P 500 constituents by a divisor. Formally, the divisor satisfies the continuity condition:

$$\sum_{i \in \mathcal{X}_t} P_{i,t} S_{i,t} / \text{Divisor}_t = SP500_t = \sum_{i \in \mathcal{X}_{t-1}} P_{i,t} S_{i,t-1} / \text{Divisor}_{t-1}, \quad (\text{A2})$$

or equivalently:

$$\text{Divisor}_t / \text{Divisor}_{t-1} = \sum_{i \in \mathcal{X}_t} P_{i,t} S_{i,t} / \sum_{i \in \mathcal{X}_{t-1}} P_{i,t} S_{i,t-1}. \quad (\text{A3})$$

The divisor ensures that the S&P 500 index level is unaffected by corporate actions such as changes in shares outstanding, special dividends, or constituent changes. Economically, this means the index behaves like a portfolio that automatically reinvests cash flows from share repurchases and special dividends; ordinary dividends therefore become the only cash flow accruing to investors. Accordingly, De La O and Myers (2021) restrict their dividend measure to ordinary dividends, excluding special items and share repurchases. Since the divisor is not publicly disclosed, it can be recovered as the ratio of total market capitalization to the index level:

$$\widehat{\text{Divisor}}_t = M_t / SP500_t, \quad (\text{A4})$$

where total market capitalization M_t is obtained from CRSP and the index level $SP500_t$ from Yahoo Finance. The aggregate dividend index is then defined as:

$$\text{Div}_t = D_t / \widehat{\text{Divisor}}_t. \quad (\text{A5})$$

The aggregate earnings index is constructed following the same procedure, with earnings per share replacing dividends per share as the underlying firm-level variable. Quarterly dividend and earnings growth rates are computed as the log difference between the index level four quarters ahead and the current level, as illustrated for dividends below. Because the analysis is conducted at the index level, aggregate dividend and earnings series are strictly positive throughout the sample. This ensures that log differences are always well-defined:

$$\Delta d_{t+1} = \log(\text{Div}_{t+1}) - \log(\text{Div}_t) = \log\left(\frac{\text{Div}_{t+1}}{\text{Div}_t}\right). \quad (\text{A6})$$

Appendix 2: Validation of the Realized Index

Table A1 validates the aggregated realized dividend and earnings indices (*All Companies*) against the benchmarks used by De La O and Myers (2021): Shiller’s historical series (*Shiller*), the SPY ETF payouts (*SPY*), and an aggregated realized dividend and earnings series constructed using only forecast-covered firms, scaled by the ratio of total to covered market capitalization (*I/B/E/S*). To allow direct comparison with their reported figures, the analysis of Table 1 is repeated for the original sample, ending in 2015Q3 and referred to as the replication subsample.

Table A1: Correlations of S&P 500 Cash Flow Measures – Replication Subsample

	Levels			Growth		
	I/B/E/S (1)	Shiller (2)	SPY (3)	I/B/E/S (4)	Shiller (5)	SPY (6)
<i>Panel A: Dividends – 2004Q1–2015Q3 (N = 47)</i>						
All Companies	0.998	0.994	0.989	0.971	0.966	0.935
I/B/E/S		0.995	0.992		0.951	0.915
Shiller			0.998			0.972
<i>Panel B: Earnings – 1977Q1–2015Q3 (N = 155)</i>						
All Companies	0.994	0.993		0.992	0.923	
I/B/E/S		0.993			0.916	

Note: The table reports pairwise correlations for quarterly S&P 500 cash flow series, restricted to the replication subsample of De La O and Myers (2021) ending in 2015Q3. *All Companies* aggregates realized cash flows across all index constituents; *I/B/E/S* aggregates realized cash flows only for constituents with analyst coverage, rescaled by the market-capitalization ratio; *Shiller* refers to the quarterly S&P 500 series from Shiller (2015). Panel A additionally benchmarks against SPY (SPDR S&P 500 ETF, CRSP). Panel B uses three series only (no SPY earnings equivalent). The growth columns report correlations of log annual changes.

For dividends, the *All Companies* series exhibits level correlations of 0.994 and 0.989 with the *Shiller* series and *SPY*, respectively. The corresponding figures reported by De La O and Myers (2021) are 0.995 and 0.994. In terms of one-year ahead log growth rates, the replicated

correlations are 0.966 and 0.935, compared to 0.937 and 0.928 in the original. The differences are negligible, confirming that the dividend aggregation is successfully replicated.

For earnings, the *All Companies* series yields a level correlation of 0.993 with the *Shiller* series, compared to 0.994 in the original. The growth rate correlation is 0.923, compared to 0.968 in De La O and Myers (2021). While the deviation in growth rates is slightly larger, all correlations remain high, and the replication closely reproduces the original results across dividends and earnings and in levels and growth rates.

Figure A1 provides complementary visual confirmation, covering the replication subsample and extending it through 2023Q3. All series track each other closely, and the tight co-movement visible in the figure is consistent with the high correlations reported in Table A1.

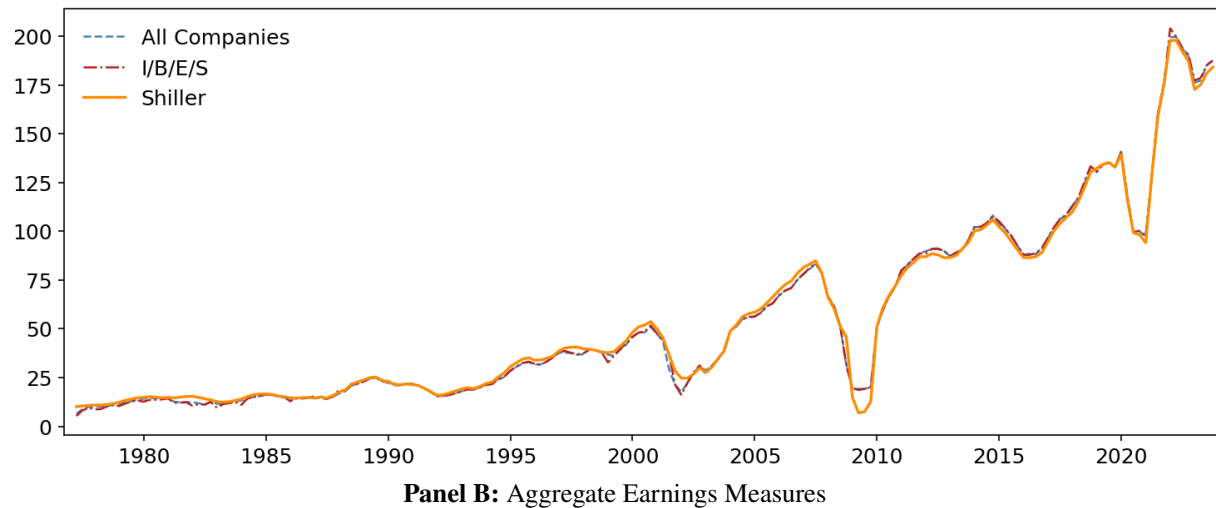
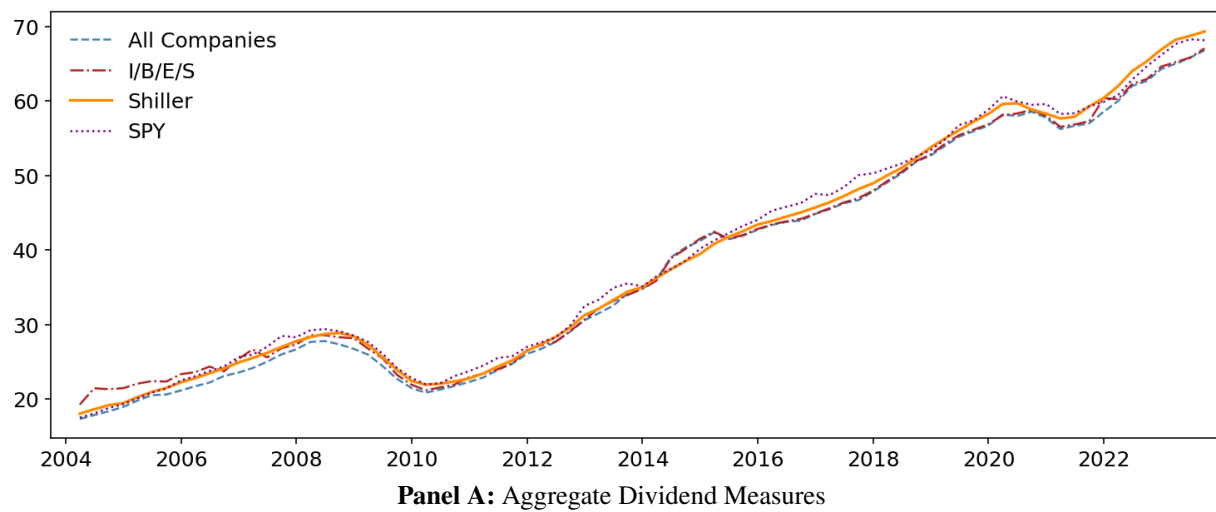


Figure A1: Validation of Aggregated S&P 500 Cash Flow Indices

Note: Panel A plots four quarterly measures of aggregate S&P 500 dividends. Panel B plots three quarterly measures of aggregate S&P 500 earnings. In both panels, the dashed line (*All Companies*) covers all index constituents and the dash-dotted line (*I/B/E/S*) is restricted to constituents with analyst coverage, rescaled by the ratio of total to covered market capitalization. The solid line (*Shiller*) is taken from Shiller (2015). In Panel A, the dotted line (*SPY*) represents trailing four-quarter dividends of the SPDR S&P 500 ETF. Panel A covers the period from 2004Q1 to 2023Q3 and Panel B covers the period from 1977Q1 to 2023Q3.

Appendix 3: Constructing the Short-Term Expectations Index

For the expectations measure, firm-level forecasts of dividends per share are obtained from I/B/E/S and are linearly interpolated to yield exact one- and two-year ahead forecasts, as described in Section 2. Then, the same aggregation procedure as for the realized index is applied, summing across covered firms and dividing by the index divisor $\widehat{\text{Divisor}}_t$. This yields:

$$E_t^*[\text{Div}_{t+1}] = E_t^* \left[\sum_{i \in \mathcal{X}_{t+1}} D_{i,t+1} S_{i,t+1} / \widehat{\text{Divisor}}_{t+1} \right]. \quad (\text{A7})$$

One issue must be addressed, however: The expression above contains future constituents, shares outstanding, and divisor values that are unknown at the time the forecast is made. To resolve this, De La O and Myers (2021) introduce the following assumption.

[Assumption 1]

$$E_t^* \left[\frac{\sum_{i \in \mathcal{X}_{t+1}} D_{i,t+1} S_{i,t+1}}{\sum_{i \in \mathcal{X}_t} D_{i,t+1} S_{i,t}} \right] = E_t^* \left[\frac{\sum_{i \in \mathcal{X}_{t+1}} P_{i,t+1} S_{i,t+1}}{\sum_{i \in \mathcal{X}_t} P_{i,t+1} S_{i,t}} \right].$$

Investors expect any future changes in constituents or shares outstanding to affect total dividends and total market value in the same proportion. This leaves the price-dividend ratio of the index unchanged, consistent with the mechanical role of the divisor. Under this assumption, the unobservable future constituent set and the corresponding shares outstanding can be replaced by their current counterparts, and the expected aggregate dividend simplifies to:

$$E_t^* \left[\sum_{i \in \mathcal{X}_{t+1}} D_{i,t+1} S_{i,t+1} / \widehat{\text{Divisor}}_{t+1} \right] = E_t^* \left[\sum_{i \in \mathcal{X}_t} D_{i,t+1} S_{i,t} / \widehat{\text{Divisor}}_t \right] = \sum_{i \in \mathcal{X}_t} E_t^*[D_{i,t+1}] S_{i,t} / \widehat{\text{Divisor}}_t. \quad (\text{A8})$$

[Assumption 2]

$$\frac{\sum_{i \in \mathcal{X}_t} E_t^*[D_{i,t+1}] S_{i,t}}{\sum_{i \in \mathcal{X}_t^j} E_t^*[D_{i,t+1}] S_{i,t}} = \frac{M_t}{M_t^j}, \quad \text{where } M_t^j = \sum_{i \in \mathcal{X}_t^j} P_{i,t} S_{i,t}.$$

Furthermore, I/B/E/S does not provide forecasts for all S&P 500 firms because some companies do not have analyst coverage. An adjustment for missing observations is necessary. Following De La O and Myers (2021), aggregate expectations are scaled by the ratio of total S&P 500 market capitalization M_t to the market capitalization of covered firms M_t^j , where $\mathcal{X}_t^j \subseteq \mathcal{X}_t$ denotes the subset of firms with available forecasts for horizon j . This scaling is based on the assumption that covered firms constitute a representative sample of the full index.

The aggregate expected dividend measure then becomes:

$$E_t^*[\text{Div}_{t+1}] = \frac{M_t}{M_t^j} \sum_{i \in \mathcal{X}_t^j} E_t^*[D_{i,t+1}] S_{i,t} / \widehat{\text{Divisor}_t}. \quad (\text{A9})$$

Figure A1 provides visual evidence that the representativeness assumption underlying Assumption 2 is valid: The *I/B/E/S* subsamples of realized earnings and dividend indices track the full-sample realized indices (*All Companies*) and the external benchmark series closely over the entire sample period. This close co-movement is confirmed quantitatively by the correlation figures in Table A1. For dividends, the *I/B/E/S* series exhibits level correlations of 0.998, 0.995, and 0.992 with the *All Companies* series, the *Shiller* benchmark, and *SPY*, respectively, closely in line with the 0.996, 0.995, and 0.993 reported by De La O and Myers (2021). Growth rate correlations of 0.971, 0.951, and 0.915 are likewise consistent with their figures of 0.974, 0.901, and 0.892. For earnings, the *I/B/E/S* series yields level correlations of 0.994 and 0.993 with the *All Companies* series and the *Shiller* benchmark, compared to 0.999 and 0.990 in De La O and Myers (2021), and growth correlations of 0.992 and 0.916 against their 0.991 and 0.946. Together, the high correlations across all measures, sample periods, and benchmarks confirm that the scaling adjustment successfully recovers the behavior of the full index from the covered subset.

Finally, expected cash flow growth rates are derived from the constructed level forecasts. Since *I/B/E/S* forecasts are expressed in levels and not in logs, one-year expected dividend growth is approximated as the log difference between the one-year ahead forecast level and the current realized value:

$$E_t^*[\Delta d_{t+1}] \approx \log(E_t^*[\text{Div}_{t+1}]) - \log(\text{Div}_t). \quad (\text{A10})$$

Two-year expected dividend levels are constructed using the same aggregation procedure, and two-year expected growth is defined as the log difference between the two-year and one-year expected levels, capturing the incremental growth expected in the second year:

$$E_t^*[\Delta d_{t+2}] \approx \log(E_t^*[\text{Div}_{t+2}]) - \log(E_t^*[\text{Div}_{t+1}]). \quad (\text{A11})$$

All steps are applied analogously to construct one- and two-year earnings level forecasts and the corresponding earnings growth expectations.

Appendix 4: Constructing the Long-Term Expectations Index

Complementing the short-term dividend and earnings expectation indices constructed above, the long-term expectations measure is considered. Following Bordalo et al. (2024), this paper

constructs an aggregate long-term growth (LTG) forecast index for earnings and dividends, capturing investor expectations over a horizon of three to five years. Firm-level forecasts are aggregated to the index level by constructing a value-weighted average:

$$\text{LTG}_t = \sum_{i \in \mathcal{X}_t} \text{LTG}_{i,t} \cdot \frac{P_{i,t} \cdot S_{i,t}}{\sum_{i \in \mathcal{X}_t} P_{i,t} \cdot S_{i,t}}. \quad (\text{A12})$$

Value-weighting ensures that larger firms, which contribute more to the overall index, receive a proportionally greater weight in the aggregate forecast.

Appendix 5: I/B/E/S Forecast Coverage for DPS

Figure A2 documents the forecast coverage underlying the constructed indices, which varies considerably across horizons and forecast types. EPS LTG coverage serves as the reference series, which remains stable at approximately 500 constituents from 1983 onward. Turning to DPS coverage, the 1-year and 2-year series track the EPS LTG reference closely and largely overlap after 2006. DPS LTG coverage, on the other hand, rises from near zero before 2003 to a peak of roughly 200 to 300 firms around 2007 and 2012, before collapsing after 2015 with fewer than five constituents reporting thereafter. This reflects the discontinuation of DPS LTG data collection by I/B/E/S. The DPS LTG series is therefore restricted to 2003Q1–2015Q3 across the entire analysis, corresponding to the replication subsample of De La O and Myers (2021).

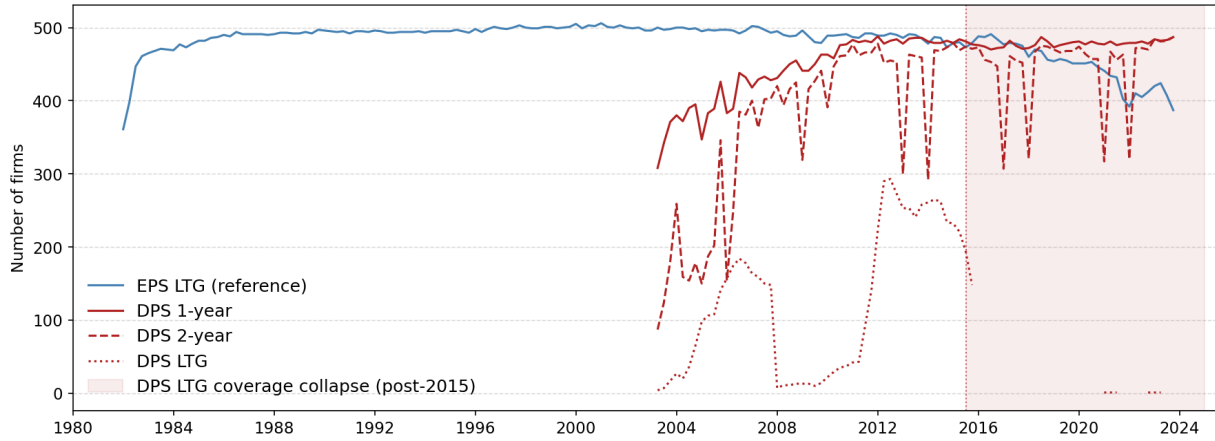


Figure A2: S&P 500 I/B/E/S Forecast Coverage by Horizon: DPS and EPS LTG

Note: This figure plots the number of S&P 500 constituents with a valid I/B/E/S consensus forecast per quarter. The solid blue line shows EPS long-term growth (LTG) coverage as a reference series. Red lines show DPS coverage at the 1-year ahead (solid), 2-year ahead (dashed), and long-term growth (dotted) horizons. The shaded region indicates the post-2015Q3 period in which DPS LTG coverage collapses to near zero.

Zusammenfassung in deutscher Sprache

Diese Arbeit untersucht, welchen Beitrag Analystenerwartungen unterschiedlicher Prognosehorizonte zur Erklärung von Bewertungskennzahlen des S&P 500 und zur Vorhersage aggregierter Aktienrenditen leisten. Ausgangspunkt ist das Barwertmodell, in dem Bewegungen von Aktienpreisen entweder auf Änderungen erwarteter zukünftiger Cashflows oder auf Änderungen erwarteter Renditen zurückzuführen sind (De La O & Myers, 2021). Auf Basis vierteljährlicher Daten aus CRSP, Compustat, I/B/E/S und der Graham-Harvey-Umfrage zum globalen Geschäftsausblick repliziert und erweitert diese Arbeit die subjektive Varianzzerlegung von De La O und Myers (2021) bis 2023Q3, untersucht die Renditevorhersagbarkeit von Gewinnerwartungen über verschiedene Horizonte in Anlehnung an Bordalo et al. (2024) und testet mithilfe von Coibion-Gorodnichenko-Regressionen, ob langfristige Erwartungen systematische Verzerrungen aufweisen (Coibion & Gorodnichenko, 2012, 2015).

Die Ergebnisse sind horizontabhängig. Kurzfristige Cashflow-Erwartungen erklären einen großen Teil der Schwankungen von Bewertungskennzahlen, insbesondere des Kurs-Gewinn-Verhältnisses. Im verlängerten Stichprobenzeitraum bleibt dieser Befund bestehen, aber seine Stärke nimmt gegenüber dem Zeitraum 2003Q1–2015Q3 ab. Der kurzfristige Cashflow-Anteil beim Kurs-Gewinn-Verhältnis sinkt von 93,5% auf 77,5% und beim Kurs-Dividenden-Verhältnis von 39,4% auf 19,6%. Damit bestätigt die Arbeit zwar die zentrale Aussage von De La O und Myers (2021), zeigt jedoch auch, dass deren Ergebnisse teilweise vom kurzen und besonders volatilen Ausgangszeitraum geprägt sind.

Zusätzlich sprechen die Ergebnisse dafür, dass kurzfristige Gewinnerwartungen vor allem vorübergehende Gewinndynamiken abbilden und daher zwar Bewertungskennzahlen erklären, aber nicht zukünftige Renditen. Diese Interpretation steht im Einklang mit der Literatur, die betont, dass kurzfristige Erwartungen häufig den Nenner des Kurs-Gewinn-Verhältnisses nachzeichnen, anstatt anhaltende Preisbewegungen zu erfassen (Adam & Nagel, 2022; Bordalo et al., 2024; Nagel, 2024). Im Gegensatz dazu besitzen langfristige Gewinnerwartungen (LTG) eine deutlich stärkere Vorhersagekraft für aggregierte Renditen: Hohe langfristige Wachstumserwartungen sagen künftige Renditen über Drei- und Fünfjahreshorizonte signifikant negativ voraus, während einjährige Gewinnerwartungen kaum systematische Vorhersagekraft aufweisen. Dieses Muster entspricht den Befunden von Bordalo et al. (2019) und Bordalo et al. (2024), wonach überoptimistische langfristige Erwartungen vorübergehend überhöhte Bewertungen erzeugen, die sich später korrigieren und zu niedrigen Renditen führen.

Diese Arbeit wendet Coibion-Gorodnichenko-Regressionen auf aggregierte LTG-Revisionen an, um zu prüfen, ob diese langfristigen Erwartungen tatsächlich verzerrt sind. Die Ergebnisse zeigen, dass auf positive Revisionen langfristiger Gewinnerwartungen realisierte Gewinne folgen, die hinter den revidierten Erwartungen zurückbleiben. Dieses Ergebnis spricht für eine Überreaktion in langfristigen Erwartungen und ist vereinbar mit der Literatur zu Erwartungsverzerrungen bei

langfristigen Analystenprognosen (Bastianello, 2025; Bordalo et al., 2019, 2024). Insgesamt ergibt sich folgendes Bild: Kurzfristige Erwartungen erklären einen erheblichen Teil der Variation von Bewertungskennzahlen, weisen jedoch kaum robuste Vorhersagekraft für zukünftige Renditen auf. Langfristige Erwartungen dagegen bewegen sich nur schwach mit Bewertungskennzahlen, hängen jedoch negativ mit zukünftigen Renditen über längere Horizonte zusammen und deuten auf systematische Überreaktion hin. Die Ergebnisse sind mit der Sichtweise vereinbar, dass kurzfristige Erwartungen vor allem aktuelle Gewinnentwicklungen widerspiegeln, während langfristige Erwartungen stärker mit verzerrten Überzeugungen zusammenhängen, die Preise vorübergehend über fundamentale Werte hinaustreiben (Adam & Nagel, 2022; Bordalo et al., 2019, 2024).

Eine zentrale Einschränkung bleibt aber bestehen. Mit den verfügbaren Daten lässt sich nicht abschließend klären, ob LTG tatsächlich subjektive Erwartungen misst oder ob Analysten ihre langfristigen Wachstumsprognosen aus aktuellen Preisen ableiten und damit zeitvariierende erwartete Renditen widerspiegeln (Nagel, 2024). Direkte Befragungsdaten auf Investorenebene sowie eine Untersuchung der individuellen Prognosebildung von Analysten bieten Ansatzpunkte, um diese Frage in künftigen Arbeiten abschließend zu beantworten.

Declaration of Originality

I herewith declare that I have composed the present thesis myself and without use of any other than the cited sources and aids. Quotations, whether sentences or parts of sentences, are marked as such. Visualisations and other imitations are identified in terms of their message and scope with reference to the source. This thesis has not been submitted to any examination body in the same or similar form, nor has it been published. It has not been used in part or in full in any other examination or coursework. I also confirm that the written (bound) versions of this thesis submitted by me are identical in content to the electronic version submitted. Furthermore, I am aware that the use of machine-generated content does not guarantee the quality of content and text. I certify that I have used generating Artificial intelligence tools (AI tools) solely as aids and that the creative influence predominates in this work. In addition, I assure that all text passages, visualisations, data generation, data transformations and any other content-related contribution that was written with the help of AI-supported programs have been marked accordingly and provided with a reference to the AI-supported program used. I confirm that I have not used any AI-supported programmes whose use has been demonstrably excluded by the examiner. I have taken note of the 'Guidelines for students on the use of AI-supported programmes'.

Date, Signature: 25.03.2026 Anna Rapp